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Global projections of plastic use, end-of-life fate and potential changes in consumption, reduction, recycling and replacement with bioplastics to 2050

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ABSTRACT

Excessive production, indiscriminate consumption, and improper disposal of plastics have led to plastic pollution and its hazardous environmental effects. Various approaches to tackle the challenges of reducing the plastic footprint have been developed and applied, such as the production of alternative materials (design for recycling), the production and use of biodegradable plastic and plastics from power-to-X, and the development of recycling approaches. This study proposes an optimisation strategy based on regression to evaluate and predict plastic use and end-of-life fate in the future based on historical trends. The mathematical model is formulated and correlations based on functions of time are developed and optimised by minimising the sum of squared residuals. The plastic quantities up to the year 2050 are projected based on historical trends analysis, and for improved sustainability, projections are additionally based on intervention analyses. The results show that the global use of plastics is expected to increase from 464 Mt in 2020 up to 884 Mt in 2050, with up to 4725 Mt of plastics accumulated in stock in 2050 (from the year 2000). Compared to other available forecasts, a slightly lower level of plastic use and stock are obtained. The intervention analysis estimates a range of global plastics' consumption between 594 Mt and 1018 Mt in 2050 by taking into account its different increment rates (between −1 % and 2.65 %). In the packaging sector, the implementation of reduction targets (15 % reduction in 2040 compared to 2018) could lead to a 27.3 % decrease in plastic use in 2050 as compared to 2018, while achieving recycling targets (55 % in 2030) would recycle >75 % of plastic packaging in 2050. The partial substitution of fossil-based plastics with bioplastics (polyethylene) will require significant land area, between 0.2×10^6 km^2 for obtaining switchgrass and up to around 1.0×10^6 km^2 for obtaining forest residue (annual yields of 58.15 t/ha and 3.5 t/ha) in 2050. The intervention analysis shows that proactive policies can mitigate sustainability challenges, however achieving broader sustainability goals also requires reduction of footprints related to energy production and virgin plastic production, the production of bio-based plastics, and the full implementation of recycling initiatives.

1. Introduction

Extensive use of plastics for different applications has led to the present era being known as the plastic age (Thompson et al., 2009). Plastics have brought significant economic benefits, and, especially since 1950, plastic production has increased exponentially with no signs of slowing down (Abrahms-Kavunenko, 2023). Around half of all plastics produced have been made in the last 15 years (Allen et al., 2020), and their production is projected to increase by 2–3 times by 2050 (Gould, 2022). By 2050, global plastic production is estimated to reach between 902 Mt (Geyer et al., 2017) to 1124 Mt (Ellen MacArthur Foundation et al., 2017).

The use of plastics is due to its desirable properties of being

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Nomenclature

Abbreviations

ABS	Acrylonitrile butadiene styrene
ANN	Artificial Neural Network
ARIMA	Autoregressive integrated moving average
ASA	Acrylonitrile styrene acrylate
DNN	Deep Neural Network
EoL	End-of-life
GAMS	General algebraic modelling system
GDP	Gross domestic product
GHG	Greenhouse gas
HDPE	High-density polyethylene
LDPE	Low-density polyethylene
LLDPE	Linear low-density polyethylene
LSTM	Long Short-Term Memory
NLP	Nonlinear programming
OECD	Organisation for Economic Cooperation and Development
PBT	Polybutylene terephthalate
PC	Polycarbonate
PET	Polyethylene terephthalate
PMMA	Poly(methyl methacrylate)
PP	Polypropylene
PS	Polystyrene
PTFE	Polytetrafluoroethylene
PUR	Polyurethane
PVC	Polyvinyl chloride
SAN	Styrene acrylonitrile resin
SSR	Sum of squared residuals
SVM	Support vector machine
XGBoost	Extreme gradient boosting

Variables

C_n, $n \in \{1, 2, \ldots, 42\}$	coefficients for correlation
$q_{m,\text{application}}$	plastic use by application (Mt/y)
$q_{m,\text{EoL}}$	end-of-life of plastic waste (Mt/y)
$q_{m,\text{region}}$	plastic use by region (Mt/y)
Y	time (year)
$q_{m,\text{type}}$	plastic use by polymer type (Mt/y)

Parameters

MAD	mean absolute deviation
R^2	correlation of determination
RMSE	root mean square error
SST	total sum of square

inexpensive, light-weight, and durable. Plastics can reduce fuel consumption in transport (British Plastic Federation, 2023) and reduce food waste generation by keeping food fresh longer (Ellen MacArthur Foundation et al., 2017). The increase in plastic waste is driven mostly by products with short lifespans, such as packaging and single-use plastics (OECD, 2022b). Around 40 % of global plastic production is used for packaging, making packaging the largest plastic production market (Geyer et al., 2017) and the logical target sector for mitigation. Single-use plastic materials are also necessary for healthcare applications such as personal protective equipment, syringes, sterile packaging for instruments, life support equipment, etc. (North and Halden, 2013). Plastic materials are also used in surgery for joint replacement devices, stents, and other products that are integrated into the human body, such as pacemakers (Naik, 2017). Compared to metal products in medicine, plastic materials have the advantage of being lighter in weight, they can be moulded more ergonomically to fit the needs of an individual, and can be produced at a lower cost (Reed, 2022).

Plastic use for health and medical applications is unpredictable and could increase during medical disasters such as disease outbreaks (Teymourian et al., 2021). Even before the outbreak of the COVID-19 pandemic, the management of plastic waste was considered a priority environmental issue due to its close link with pollution of the terrestrial and marine environment. The widespread use of protective equipment worldwide leads to serious waste management problems and inappropriate disposal methods. Apart from the more frequent use of personal protective equipment, perishable and takeaway food should be wrapped in plastics during lockdowns and home quarantines, which leads to increased amounts of single-use plastics (Mallick et al., 2021). >90 % of plastic is petroleum-based and non-biodegradable (Zhao et al., 2020), suggesting a pathway for enhancing the environmental sustainability of plastic production.

Plastic pollution is a pressing global environmental challenge (Gould, 2022), as most plastics end up in landfills, incineration plants, or are disposed of improperly. The widespread use and unethical disposal of plastics result in plastics polluting the terrestrial and aquatic systems (Geyer et al., 2017). Smaller plastic particles (micro, nanoplastics) may have adverse health effects (Prata et al., 2020). Even with immediate and concerted action, it is estimated that 710 Mt of plastic waste will enter aquatic and terrestrial ecosystems around 2040 (Lau et al., 2020). Furthermore, greenhouse gas (GHG) emissions occur at every stage of the plastic life cycle, contributing a significant share of the total emissions (Shen et al., 2020). Most of plastics used for packaging are mechanically recyclable; however, only 14 % of plastic packaging is collected for recycling. They are mostly recycled into lower-value applications, and cannot be recycled again beneficially from an economic point of view (Ellen MacArthur Foundation et al., 2017).

Preventive measures against plastic pollution are urgently needed, particularly to address the escape routes of plastics (macro, micro, nano) through existing waste management systems (Hettiarachchi and Meegoda, 2023). Several countries have implemented plastic bans (Nikiema and Asiedu, 2022) to minimise the utilisation rate. Recently, there have also been calls for an international policy framework to manage plastic pollution (Raubenheimer and Urho, 2020). Table 1 shows the plastic related policies and targets issued by governments and organisations. Some of the targets remain ambitious and other mitigation solutions (Zheng and Suh, 2019) which do not depend on demand management are required.

Apart from technical solutions, addressing plastic pollution also

Table 1
Policies and programmes related to plastic use and recycling targets.

Policy	Target	Reference
A European strategy for plastics in a circular economy	All plastic packaging is recyclable by 2030 in the EU.	(European Commission, 2018b)
The Packaging Directive	By 2030, 55 % of plastic packaging is recycled in the EU.	(European Commission, 2022)
ReShaping Plastics	60 % less waste is incinerated between 2020 and 2050. Less (>70 %) virgin plastic is produced from fossil fuels.	(Plastics Europe, 2022b)
The Global Commitment	100 % of plastic packaging is reusable, recyclable, or compostable by 2025.	(Ellen MacArthur Foundation, 2022)
The Regional Action policy scenario	Global plastic use to decrease by ~20 % (to 1018 Mt) in 2060.	(OECD, 2022d)
The Global Ambition policy scenario	Increased demand for plastic can be met by recycled secondary plastic.	(OECD, 2022a)

requires understanding and influencing human behaviour. A comprehensive literature review found that despite a high level of awareness of the issues involved, people continue to value and routinely use plastic. Habits, norms, and situational factors are strong predictors of plastic consumption behaviour. This highlights the importance of integrating policy and psychological interventions to change consumption patterns, even if their long-term effectiveness often remains uncertain. Projections can be used to raise public awareness of the long-term impacts of plastic waste. This can lead to behavioural changes that result in more sustainable consumption patterns and support the measures to reduce plastic consumption (Heidbreder et al., 2019). For example, the effectiveness of behaviour change strategies has been studied in different contexts and for different behaviours. It included different intervention categories such as i.) informational (e.g., environmental messaging or awareness raising), ii.) visual prompts (e.g., signage and labelling), iii.) contextual (e.g., financial incentives or provision of alternatives), iv.) social (e.g., direct supervision or citizen science), and v.) personal (e.g., commitment and self-monitoring) (Nuojua et al., 2024).

Tackling the hazards posed by plastics requires multiple actions from different stakeholders (Jia et al., 2019). To design more effective policies, a better understanding of patterns of plastic use and waste generation trends is needed. In addition, understanding and predicting trends in plastic consumption and waste generation can support progress towards achieving various Sustainable Development Goals (SDGs), particularly those focused on environmental sustainability, health, and economic growth. By forecasting plastic waste trends, health risks can be anticipated, leading to preventive measures and policies to protect public health (SDG3), while the identification of future hotspots of plastic pollution can help in prioritising areas for intervention and improving waste management practices to protect water quality (SDG6) and reduce marine pollution and protect marine biodiversity (SDG14) (de Sousa, 2021). Urban planning can incorporate data on future plastic waste to design efficient waste management systems, promote recycling, and reduce landfill use (SDG11), and the effective land management policies can support conservation efforts (SDG15). Forecasting can guide the development of circular economy strategies, encouraging recycling, reuse, and reduced plastic production (SDG12) (Elsheekh et al., 2021).

Despite considerable efforts to predict plastic production, waste generation and environmental impacts, there are still some research gaps. A key challenge is the selection of representative predictors, especially in global-level analyses where complex variables interact across regions and sectors. In addition, many time series models applied in previous studies (see section 2 Literature review) are not flexible enough to capture non-linear dynamics and the full range of patterns that can influence plastic production and waste projections.

To address this research gap in identifying representative predictors, this study introduces a regression-based optimisation approach grounded in a single predictor, time, to forecast future data on plastics. The correlations as functions of time are obtained to fit the provided data and optimised by minimising the sum of squared errors. By analysing historical data, projections are made about future trends that follow the identified patterns or additionally take into account possible policy measures and regulations that could influence plastic consumption until 2050. By taking these measures into account, the projections provide insights into possible future trends and thus they could be used as a decision-making tool. The strength of this approach lies at its simplicity which could lead to better interpretability as the output is directly related to the historical data. The application of historical data for prediction could be supported by the concept of waves of innovation (Hargroves and Smith, 2013). The model could capture historical trends, seasonal patterns, and any other regularities present in the time series data, helping to make more informed projections for the future. It may also be more robust to missing data on potential predictors, particularly for global scale prediction, and can avoid the potential issues of overfitting when too many predictors are used, as it captures mainly the underlying trends in time series data. However, as for every method, varying degrees of trade-offs in accuracy and limitations in interpreting the results exist. The model performance is assessed using the correlation of determination, mean absolute deviation, and root mean square error.

This study conducts projections of plastic consumption in different regions, plastic use by polymer types and by application, and plastic waste by end-of-life (EoL) fate on a global basis and at the region level up to 2050. Apart from the analysis based on historical trends, projections up to 2050 are developed based also on the following three intervention analyses aimed at mitigating the plastic pollution and reducing plastic demand:

- The different rates of decrease/increase in plastic consumption are examined considering the growth rate of plastic volumes in 2050 between −1 % and 2.65 %.
- For the plastics used in packaging, taking into account reduction and recycling targets, it is considered that 55 % plastic packaging should be recycled by 2030 and plastic packaging volumes should be reduced by 15 % in 2030 as compared to the volumes in 2018.
- Partial substitution of fossil-based plastics with bio-based plastics is considered, with a focus on polyethylene (PE).

The novel contributions of this work include an optimisation approach-based regression model which has good interpretability without significant loss of accuracy, less sensitivity to outliers and missing data of predictor variables. Notably, it employs an expanded regression function with various terms capable of capturing patterns while maintaining simplicity as it does not rely on additional indicators. This study predicts plastic use by region and polymer type, and EoL fate utilising a different methodology and is able to complement the existing literature results, alerting to a substantial increase of plastic use and waste, including in the case of less effective policy actions.

2. Literature review

There have been various studies conducted in forecasting and projecting the plastic waste, or municipal solid waste, in general. The approaches range from statistical analysis models to more advanced methods based on machine learning. Each method has its strengths and limitations. They also slightly differ in the main assessment purpose, where machine learning generally sacrifices interpretability for predictive power and statistical analysis focuses on assessing the significance of relationships between variables while also catering for prediction (Stewart, 2019). The strengths of statistical modelling are in enabling informed interpretations about the relationships and identifying key drivers, thereby providing an understanding of the underlying dynamics. Machine learning has been found to outperform classical statistical methods when predicting the amount of waste (Garre et al., 2020). Although machine learning methods are often associated with limited interpretability and are considered as black box approaches, it should be noted that there has been a growing effort to explore the underlying relationships between model attributes and to enhance interpretability. Both approaches are equally significant and serve their own purpose for projecting and forecasting. These remain as important approaches for policy implementation and decision-making.

Various prediction approaches and forecasting studies regarding waste generation have been reported. Among the studies, the future of plastics (production, waste, and use by economic sector) was projected using a combination of historical data and empirical relationships, applying assumptions and inputs about future development (Stegmann et al., 2022). It takes the strength of statistical/empirical-based projection to assess the potential CO_2 mitigation. Time series analysis for waste prediction has also been discussed (Ghinea et al., 2016), which includes the linear trend model, quadratic trend model, exponential growth curve, and S-curve trend model. It is highlighted that trend analysis for different waste fractions (e.g., paper, glass, plastic, metals, biodegradable, and other wastes) evaluated by measuring mean absolute percentage error, mean absolute deviation, and mean squared deviation, shows different accuracies of the models in their predictions. The mentioned time series models have a lower flexibility in capturing various time series dynamics and nonlinear patterns, for example, compared to the autoregressive integrated moving average method (ARIMA) (Lakhouit et al., 2023). In addition to time series models such as ARIMA, several machine learning algorithms including linear regression, regression trees, Gaussian process regression, and support vector machine (SVM) have been employed and found to be effective in accurately predicting municipal solid waste quantities.

Different machine learning tools have been widely used for predicting plastic waste properties for their potential transformation into value-added products, thereby mitigating environmental concerns associated with their disposal. Five machine learning techniques have been used to analyse and predict properties of plastic-based composite construction material, i.e., random forest, Decision Tree, Ridge, Lasso and Linear regression (Jain et al., 2023). Other methods have also been used, such as a deep learning Long Short-Term Memory (LSTM) approach was used to forecast waste generation rates based on historical data (Cubillos, 2020). Deep Neural Network (DNN) was used to forecast COVID-19 pandemic plastic pollution to support the efficient management of the increased waste volumes generated during the pandemic period (Nanehkaran et al., 2023). Another study predicted plastic waste generation by defining several predictors for the machine learning based Artificial Neural Network (ANN) model coupled with SHapley Additive exPlanations to facilitate interpretation (Fan et al., 2022). ANN was identified as the model with the best prediction accuracy compared to SVM and random forest for plastic waste prediction (Kumar et al., 2018). Commonly applied supervised machine learning approaches include support vector machine, random forest, and extreme gradient boosting (XGBoost). XGBoost is proposed as the effective method for municipal solid waste prediction (Zhang et al., 2022) with gross domestic product (GDP) and population stated as the dominant factors for prediction. However, it should be noted that the method is highly dependent on the data availability of predictors needed to train the model. The available predictors may vary from one country to another.

Defining the representative predictors is one of the challenges especially for global-scale analysis. Most of the assessments in projecting or forecasting plastic waste generation minimised the number of predictors and utilised a relatively straightforward approach for better insight. Some of these studies focus on the engineering models that describe the life cycle of plastic products, while others such as Organisation for Economic Cooperation and Development (OECD) examine the direct links between population or GDP and plastic waste (OECD, 2022c). For example, the projections are made based on the population and GDP (Lebreton and Andrady, 2019), on historical trends (Geyer et al., 2017), and on economic growth, technology change, structural change, and population growth (OECD, 2022c). However, there is a notable gap in models that integrate the specifics of plastic products into a comprehensive framework that could represent the diverse range of plastic polymers and their various economic applications. Such models would provide a detailed understanding of the relationship between economic activities and plastic usage. Additionally, existing literature does not provide a comprehensive overview of the effectiveness of key strategies that decision-makers can employ to limit plastic usage.

To address the pervasive problem of plastic pollution requires a multifaceted approach that includes regulatory, economic, psychological, social, and technological interventions. Countries have used a range of policy instruments, from outright bans on certain plastic items to economic measures such as fees, charges or taxes, targeting both retailers and consumers. These measures aim to reduce the use of plastic by either limiting its availability or increasing the cost of plastic products. In addition to these regulatory and economic approaches, psychological and social interventions play a crucial role. Education campaigns and improved infrastructure are essential for raising awareness and promoting behavioural change in line with the principles of recycling, reuse and waste prevention (Heidbreder et al., 2019). Technological advances, particularly in the production and use of bioplastics, advanced power-to-X plastic production technologies, and advanced recycling technologies, are another important way to reduce plastic pollution. These advances should focus on improving the efficiency of feedstock production, switching to more sustainable feedstocks, improving energy efficiency in production facilities and optimising recycling processes (Rafey and Siddiqui, 2023). In addition, future regulations should require the production of sustainable and repairable products. Along the entire plastic value chains it would be beneficial to assess the need to use certain materials, to minimise the use of materials and energy, select easily recyclable materials and develop products that are straightforward to dismantle at the end of their life cycle (Moyen Massa and Archodoulaki, 2024). Finally, it is necessary to develop the capability for periodic estimation of plastic pollution at a geographically disaggregated level to allow monitoring of compliance to a future global plastic treaty (Cottom et al., 2024). These combined strategies could represent a comprehensive approach to significantly reduce plastic waste and minimise its environmental impact.

3. Methods

The projections of plastic quantities across different categories are developed using a systematic approach that starts with a clear formulation of the problem. A regression-based optimisation method is used to model trends in plastic consumption, use and EoL based on historical and intervention analysis. By combining these steps, the methodology provides a framework for estimating future quantities of plastic consumption and waste management.

3.1. Problem formulation

The historical data on plastic quantities consumed by region, plastic used by polymer type and application, and the EoL fate of plastic waste on both a global scale and for the regions are used to forecast their quantities in future years and decades. For this purpose, the prediction functions are developed based on available historical data, taking into account observed changes and variations, and indirectly considering the various factors that have affected plastic use and waste over time. The quantities of plastic use and waste are forecasted by using historical data with no additional sustainability interventions.

However, the use of historical data could be subject to technological advances, regulations and policies, economic fluctuations, consumer behaviour, environmental concerns and certain historical events, which could influence the results. For example, innovations in waste management and the development of substitutes such as bioplastics could significantly alter historical trends. Changes in legislation, including additional taxes, bans or recycling requirements, can also affect the use of plastics and management practices. Economic conditions and market dynamics play a crucial role in influencing production and consumption patterns. In addition, shifts in consumer preferences towards more sustainable practices and a growing awareness of environmental issues can have an impact on the use and disposal of plastics. Certain historical events such as wars or pandemics can disrupt production and consumption patterns, leading to further distortions in the data.

An analysis considering different interventions towards sustainability is thus conducted in this study. The reduction of the plastic consumption is influenced by the implementation of the new regulations and policies, including taxes to discourage plastic consumption, bans on single-use plastics, and consequential consumer behaviour shifts towards sustainability. These changes aim to grow a preference for products with less packaging, a higher demand for reusable or recyclable items, and overall environmentally conscious purchasing decisions. In addition, the potential technological advances in waste management are considered to improve the recycling and processing efficiency. This includes innovations in recycling technologies and methods for converting plastic waste into reusable materials or energy. By setting mandatory recycling targets and offering financial incentives for adopting more sustainable practices, a significantly increased recycling rate of plastic materials can be achieved. The technological development of substitute materials such as bioplastics could affect their availability and increased use, offering more sustainable alternatives to conventional plastics. The advancements in developing and deploying bioplastics and other sustainable materials are considered to promote reduced reliance on fossil fuels and decreasing environmental impacts. The following projections considering more sustainable use of plastics are analysed:

- Projections of global plastic consumption under varying growth scenarios, considering different increment rates for global plastic consumption. An average growth rate is determined from the projections of plastic consumption by region (based on historical trends analysis), and the correlations of global plastic consumption considering different increment rates in 2050 are determined to simulate future trends, together with the trends considering the initiatives to reduce plastic production volumes. The rates of increase in plastic consumption in 2050 are considered to be between −1 % and 2.65 %.
- Projections of plastic use in packaging with sustainability targets, considering reduction and recycling targets. The projections of plastic use in packaging (based on historical trends analysis) are compared with the quantities used in this sector over the forecasted horizon (up to 2050) taking into account reduction targets (−15 % by 2030) (European Commission, 2018a). In addition, the quantities of recycled packaging are determined on the basis of historical data and then correlated in order to simulate the packaging amounts with or without the recycling target (55 % of plastic packaging to be recycled by 2030) (European Commission, 2018a).
- Projections of plastic use with bioplastic substitution, considering a partial substitution of fossil-based plastics by bio-based. Based on historical data on the use of bioplastics, the correlation is developed to simulate a gradual substitution of fossil-based plastics. The polymers high-density polyethylene (HDPE) and low-density polyethylene (LDPE) are replaced by bio-HDPE and bio-LDPE by 50 % and 55 % by 2050 (Brizga et al., 2020). It is assumed that bio-based polymers are produced from bioethanol, which is obtained from biomass feedstocks such as sugarcane, corn, switchgrass and forest residue. According to the estimated quantities of bioplastics used in 2050, the quantities of biomass and the corresponding land area for harvesting or cultivation are determined.

These projections aim to evaluate the potential impacts of various strategies on plastic usage and waste management, providing a comprehensive understanding of how different approaches can mitigate environmental burdens of plastics.

3.2. Description of the method

The optimisation approach based on regression is proposed to predict the future data on i) plastic consumption by region, ii) plastic use by polymer type, iii) plastic use by application, iv) the EoL fate of plastic waste on both a global scale and v) for the regions studied. To analyse and forecast the data about plastic use and waste, the optimisation approach by applying the least squares method is proposed in this study. The mathematical model was developed to provide the prediction of trends based on previous information, whereas the optimisation model was adopted from the previous work by the authors (Dokl et al., 2022). Here, the general formulation of the model for the development of correlations is used, and modified to forecast the future trends of plastic volumes using historical data.

The data about plastic consumption in all the reported regions, plastic use by type of polymers, by application, including the EoL of plastic waste used in the model, was extracted from the OECD Global Plastics Outlook (OECD, 2022b). The data regarding plastic consumption by regions ($q_{\mathrm{m,region}}$) is obtained for the time period from 1980 to 2019, the data on plastic use by type of polymer ($q_{\mathrm{m,type}}$) and by application ($q_{\mathrm{m,application}}$) is given in the time period between 1990 and 2019, and the data regarding the EoL fate (the amounts of littered, mismanaged, incinerated, recycled, and landfilled plastic waste) are available for each region studied from the time period between 2000 and 2019. The amounts considering the EoL fate of plastic waste are considered on a global scale by summing EoL data for all the locations ($\sum_{\mathrm{region}} q_{\mathrm{m,EoL,region}}$ or also $q_{\mathrm{m,EoL,global}}$), and specifically for each region ($q_{\mathrm{m,EoL,region}}$). The detailed information on the data sets used for the projections for plastic consumption, use and EoL are listed in Table 2. The data sets for the EoL categories of plastic waste in different regions, $q_{\mathrm{m,EoL,region}}$, are not listed in the table due to the extensiveness of the information. As indicated in the table, each of the 10 regions include 5 different EoL data, leading to 49 data sets in total, as one prediction function, for incinerated waste of Other Africa ($q_{\mathrm{m,incinerated,OtherAfrica}}$), is omitted due to unavailable data.

Available historical data on plastics consumption by regions (from 1980 to 2019), uses by polymer types and by applications (from 1990 to 2019), and EoL of waste by regions (from 2000 to 2019) vary across prediction models. Consequently, the resulting data point values employed in projections are 40 for plastic consumption by regions, 30 for plastic use by polymer types and by applications, and 20 for EoL of waste generated. All plastic data concerning use by regions, polymers, applications, and EoL of plastic waste are incorporated into the prediction models using symbol q_{m} in units of Mt/y (million metric tonnes per year).

Statistical techniques, including time series regression models, assume data homogeneity, aiming to recognise patterns and capture the effects of events retrospectively. In this context, variations in observed data are indirectly integrated into prediction functions for forecasting future trends. To identify patterns and forecast the plastic volumes, this study considers time as an independent variable, which describes historical trends, and other potential indicators (excluding their identification) that may be reflected in underlying patterns. By employing this approach, this study aims to develop a robust prediction model that can effectively anticipate changes and trends in plastic use and waste over time.

To describe the available data, prediction functions are developed, representing the correlation between plastic quantities and time. The correlations are formulated with different types of equation, i.e., linear, polynomial and exponential terms, which are set for $y\left(q_{\mathrm{m,region}}, q_{\mathrm{m,type}}, q_{\mathrm{m,application}}, q_{\mathrm{m,EoL,region}}\right)$ dependent on the time period $x(Y)$, where Y represents the year for which the plastic data corresponds to.

The equations are described by the coefficients C_{n} as variables to be optimised to obtain the best fit between the provided and calculated data. The prediction models are based on an equation consisting of 42 coefficients (from C_1 to C_{42}), as shown in Eq. (1):

$$\begin{aligned} q_{\mathrm{m}} = {} & C_1 + C_2 \cdot t^{C_3} + C_4 \cdot t^{C_5} + C_6 \cdot t^{C_7} + C_8 \cdot t^{C_9} + C_{10} \cdot t^{C_{11}} + C_{12} \cdot t^{C_{13}} + C_{14} \cdot t^{C_{15}} \\ & + C_{16} \cdot t^{C_{17}} + C_{18} \cdot t^{C_{19}} + C_{20} \cdot t^{C_{21}} + C_{22} \cdot t^{C_{23}} \cdot t^{C_{24}} + C_{25} \cdot exp\left(t^{C_{26}}\right) \cdot t^{C_{27}} \\ & + C_{28} \cdot t^{C_{29}} \cdot exp\left(t^{C_{30}}\right) + C_{31} \cdot exp\left(t^{C_{32}}\right) \cdot t^{C_{33}} + C_{34} \cdot t^{C_{35}} \cdot exp\left(t^{C_{36}}\right) \\ & + C_{37} \cdot t^{C_{38}} \cdot t^{C_{39}} + C_{40} \cdot exp\left(t^{C_{41}}\right) \cdot exp\left(t^{C_{42}}\right) \end{aligned} \tag{1}$$

As mentioned above, the equation is based on an optimisation model from previous study by authors (Dokl et al., 2022). In the previous study such equation was developed to fit the data of the thermophysical properties of fluids and estimate the values in regions where the data were not provided. Here, the general structure of the model is used as a basis for projecting the data on plastic quantities based on the patterns

Table 2
Information on data sets of plastic use and waste end-of-life used for projections.

Data sets[a]	Plastic consumption	Plastic use	Plastic use	Plastic waste
	By region ($q_{m,region}$)	By polymer type[b] ($q_{m,type}$)	By application ($q_{m,application}$)	By end-of-life and region ($q_{m,EoL,region}$)[e]
	Canada ($q_{m,Canada}$) China ($q_{m,China}$) EU: OECD EU, Other EU ($q_{m,EU}$) India ($q_{m,India}$) Latin America ($q_{m,LatinAmerica}$) Middle East and North Africa ($q_{m,MEast\&NAfrica}$) Other Africa ($q_{m,OtherAfrica}$) Other Asia: OECD Asia, Other non-OECD Asia ($q_{m,OtherAsia}$) Other countries: OECD Oceania, Other OECD America, OECD non-EU and Other Eurasia ($q_{m,OtherCountries}$) USA ($q_{m,USA}$) World ($q_{m,World}$)	ABS, ASA, SAN ($q_{m,ABS,ASA,SAN}$) Bioplastics ($q_{m,Bioplastics}$) Elastomers ($q_{m,Elastomers}$) Fibres ($q_{m,Fibres}$) HDPE ($q_{m,HDPE}$) LDPE, LLDPE ($q_{m,LDPE,LLDPE}$) Marine coatings[c] ($q_{m,MarineCoatings}$) Other ($q_{m,type,Other}$) PET ($q_{m,PET}$) PP ($q_{m,PP}$) PS ($q_{m,PS}$) PUR ($q_{m,PUR}$) PVC ($q_{m,PVC}$) Road marking coatings[c] ($q_{m,RoadMC}$) Total ($q_{m,Total}$)[d]	Building and construction ($q_{m,B\&C}$) Consumer and institutional products ($q_{m,C\&IP}$) Electrical/electronics ($q_{m,E\&E}$) Industrial/machinery ($q_{m,I\&M}$) Marine coatings[c] ($q_{m,MarineCoatings}$) Other ($q_{m,application,Other}$) Packaging ($q_{m,Packaging}$) Personal care ($q_{m,PersonalCare}$) Road marking coatings[c] ($q_{m,RoadMC}$) Textile sector ($q_{m,Textile}$) Transportation ($q_{m,Transportation}$) Total ($q_{m,application,Total}$)[d]	Incinerated ($q_{m,Incinerated,region}$) Landfilled ($q_{m,Landfilled,region}$) Littered ($q_{m,Littered,region}$) Mismanaged ($q_{m,Mismanaged,region}$) Recycled ($q_{m,Recycled,region}$) Total ($q_{m,TotalEoL,region}$)
Data points (years)	1980–2019	1990–2019	1990–2019	2000–2019
Number of data points	40	30	30	20

[a] All the data are obtained from Organisation for Economic Cooperation and Development (OECD) (OECD, 2022b).
[b] Abbreviations for polymer types: acrylonitrile butadiene styrene (ABS), acrylonitrile styrene acrylate (ASA), high-density polyethylene (HDPE), low-density polyethylene (LDPE), linear low-density polyethylene (LLDPE), polyethylene terephthalate (PET), polypropylene (PP), polystyrene (PS), polyurethane (PUR), polyvinyl chloride (PVC), styrene acrylonitrile resin (SAN).
[c] Historical data for categories Marine coatings and Road marking coatings are the same (OECD, 2022b), thus also the projections obtained are identical.
[d] The data for the amounts of total amounts of polymers by type and application are the same ($q_{m,Total}$)
[e] The data are available for each EoL and region simultaneously, except the data for the incinerated waste in Other Africa ($q_{m,incinerated,OtherAfrica}$) is unknown. Besides, the data for all the regions were summed and the global data for each EoL are in this way obtained ($q_{m,Incinerated,global}$, $q_{m,Landfilled,global}$, $q_{m,Littered,global}$, $q_{m,Mismanaged,global}$, $q_{m,Recycled,global}$ and $q_{m,TotalEoL,global}$).

recorded in historical data.

Eq. (1) intentionally includes repeated terms to provide significant flexibility during coefficient optimisation. By repeating certain terms, the optimised solutions can set coefficients to any value including zero or one (depending if they are part of polynomial or exponential term), allowing elimination of non-essential or insignificant terms for prediction. Therefore, by repeating certain terms and employing a diverse set of equation types, the prediction model can effectively accommodate different patterns and trends in the plastic use data, allowing for a more accurate and comprehensive representation of the underlying dynamics.

Prediction models for all forecasted volumes (see also Table 2) are derived from Eq. (1) and consist of 42 coefficients or fewer. Some of the plastic data are obtained with optimised values of all 42 coefficients, while to obtain other results, coefficients (and terms of Eq. (1)) are either eliminated through optimisation by setting the initial coefficients in those terms to zero or terms are omitted for the reasons of obtaining infeasible solutions or unrealistic results (based on authors estimation). In these cases, fewer coefficients were considered, derived from Eq. (1), to achieve the optimal fit for the given data and to obtain future plastic predictions. Four different formulated equations, each with optimal coefficient values tailored to specific plastic data sets, have been used to obtain reasonable forecasts:

- Optimised values of all 42 coefficients are considered for the plastic consumption for the region USA, the plastic use for the polymers HDPE, PET, PP, and PVC, for the plastic use by applications in Building and construction and Packaging. Additionally, such prediction functions are performed for projecting plastic quantities with sustainability constraints: increment rates of −0.5 % and −1 % for plastic use in 2050, for packaging with reduction target an recycled packaging with recycling target;
- Optimised values of 30 coefficients are considered for plastic consumed in Latin America, for plastic used in polymer category Fibres, and for the plastic use by applications in Consumer and institutional products, up to the term involving C_{24} and then followed by additional terms $C_{25}\cdot Y^{C_{26}}\cdot Y^{C_{27}} + C_{28}\cdot exp\left(Y^{C_{29}}\right)\cdot exp\left(Y^{C_{30}}\right)$;
- Optimised values of 24 coefficients are considered for projecting plastic consumption in regions Canada, China, EU, India, the Middle East and North Africa, Other Africa, Other Asia and Other countries, for plastic use of polymers ABS, ASA, SAN, Elastomers, LDPE, LLDPE, Marine coatings, Other, PS, and PUR, for plastic use in applications Other and Transportation, for all the categories concerning the EoL of plastic waste, and for the total plastic use by polymer or by application, by region, and by EoL. Projections considering intervention analysis for improved sustainability including 24 coefficients are those with increment rates of 1.325 % and 0 % and recycled packaging without recycling target. In these cases, Eq. (1) is reduced to the term involving C_{24};
- Optimised values of only 4 coefficients are considered for the use of polymer category Bioplastics, for plastic use in applications Electrical/electronics, Industrial/machinery, Personal care, and Textile sector and for substituted fossil-based plastics by bioplastics for both HDPE and LDPE. In these cases, Eq. (1) is shortened to expression $C_1 + C_2\cdot Y^{C_3}\cdot Y^{C_4}$.

During the optimisation process, the values of coefficients are optimised to best fit the available historical data and additional constraints in the case of intervention analysis. As a result, the coefficient values are assigned differently for each prediction function, ensuring that the models are tailored to capture the unique patterns and trends specific to each category of data. The optimised values of coefficients for all the correlations obtained are collected in Tables A1-A5 in the Supplementary information.

To optimise the coefficients of equations to capture the underlying trends and patterns in the historical plastic data, the sum of squared residuals (*SSR*) is considered as the key metric for evaluating function performance. The *SSR* measures the discrepancy between the actual historical values and the values predicted by the developed model. By minimising the *SSR*, a robust evaluation of how well prediction models fit the data is provided, while allowing for an assessment of their overall performance. This supports the development of more informed and reliable projections for plastic quantities over time.

The correlations are formulated as nonlinear programming (NLP) models and optimised in General Algebraic Modelling System (GAMS) (GAMS Development Corporation, 2022b) by minimising SSR between the calculated values ($\hat{y}$) and the provided historical data (y_i) for a number of data points (n) as shown in Eq. (2). Due to the complex structure of the model and the inclusion of strongly nonlinear expressions, the CONOPT solver is used, which is designed to find locally optimal solutions (GAMS Development Corporation, 2022a). Although applied solver does not guarantee global optimality, it can effectively handle the nonlinearities in the prediction model. It is also important to note that the differences between the locally optimal results obtained were generally small, indicating the presence of multiple locally optimal solutions with similar objective function values. This suggests that, although global optimality is not assured, the solutions provided are reliable in the context of the model in this study.

$$SSR = \sum_{i=1}^{n} (y_i - \hat{y})^2 \tag{2}$$

The objective function considered ensures the forecasting is aligned as closely as possible with the historical data, enhancing the reliability and accuracy of the predictions in the study. Finally, the obtained correlations with optimised coefficients are used to predict the future plastic use and waste amounts as functions of time. The predictions are made using correlations to forecast trends up to the year 2050 by using historical data.

The deviations between the data points provided and the correlations obtained are for each data set graphically presented in Figs. B1–B4 in Part B in Supplementary information. Various performance metrics are further used to assess the accuracy and reliability of the model, including the correlation of determination (R^2), the mean absolute deviation (*MAD*) and the root mean square error (*RMSE*), as described in Part C in Supplementary information. In addition, the development of correlations is discussed to estimate future plastic volumes based on historical data, taking into account the contained sampling error. Confidence intervals are used to quantify the uncertainty of these estimates, using a 95 % confidence level to create these intervals, which is explained in Part D in Supplementary information. In summary, the obtained visual representations are consistent with the obtained values of the performance metrics and uncertainty assessments. High R^2 values and low values of *MAD* and *RMSE* together with the narrow confidence intervals confirm the accuracy and reliability of the model in predicting future plastic quantities.

3.3. Assumptions and limitations

The development of the prediction models is based on various assumptions, and the limitations of the methods are acknowledged. These include general assumptions related to terminology and the specific assumptions and limitations associated with the methodology for predicting plastic quantities based on historical data and additionally taking into account more sustainable practices of plastic use and waste management in the future.

3.3.1. General assumptions

In projecting plastic quantities, it is essential to clarify terminology related to plastic volumes such as production, consumption, use and waste. Based on the literature cited in this paper that discusses plastic volumes, the authors' understanding is that "production" usually refers to the original manufacture of plastics. In contrast, "consumption" is often used as a quantitative measure and focuses on the amount and rate at which plastics is consumed, while "use" is generally associated with the activities related to plastic items and indicates where and how various plastics are used. "Waste" refers to plastic materials that have reached the end of their life cycle and are discarded. In this study, the term "production" is excluded as the approach is based on the data on the plastics used, consumed and discarded. The term "consumption" is used in a quantitative way, referring to the data on regional plastic consumption, while "use" is used to describe the specific applications and types of plastics. The term "waste" is considered after the plastics used become discarded, which is discussed further as different EoLs.

The projection functions developed are based on the data that are extracted from a single source OECD (2022b) as mentioned above. The categories of regional plastic consumption are integrated as in the model as described in Table 2. While most of the categories refer to a certain type of polymer (i.e. ABS, ASA, SAN, HDPE, LDPE, LLDPE, PET, PP, PS, PUR and PVC), the rest of the categories such as bioplastics, elastomers, fibres, marine coatings and road marking coatings capture a variety of polymers of which they consist of. Category Other includes polymers such as polybutylene terephthalate (PBT), polycarbonates (PC), poly (methyl methacrylate) (PMMA), polytetrafluoroethylene (PTFE) (OECD, 2022b).

As for the EoL of plastic waste, five categories are considered. For the recycled waste is assumed to be collected, processed and reused to produce new plastics, excluding residues from recycling treated by other methods. Incinerated waste is burned in modern industrial plants, with or without energy recovery. Landfilled waste is disposed of on land in a controlled, hygienic and environmentally sound manner. Mismanaged waste is waste that has been improperly handled, such as incineration in open pits, dumping in water or unhygienic disposal in landfills, as well as waste that is not covered by collection systems, e.g. road markings. Littered waste is disposed of by individuals or by illegal dumping and is not collected by street cleaning or other clean-up operations, with the exception of litter disposed of by other methods (OECD, 2022b).

In addition to projecting plastic volumes based on historical data, the potential impact of various potential interventions aiming at more sustainable practices is examined regarding plastic quantities, including various regulatory measures to address the challenges of plastic production and waste management. Plastic consumption in different regions is assumed to be subject to existing and additionally introduced targets aiming at addressing consumer behaviour to reduce plastic consumption and improve waste management in general.

Waste generation is closely linked to the use of plastics and varies depending on the average lifespan of individual plastic products. For example, packaging plastics have a very short lifespan compared to plastics used in construction (Geyer et al., 2017). In addition, waste generation and recycling rates are influenced by the recyclability and value of the individual polymers. In general, it is assumed that PET and HDPE have the highest recycling rates, followed by LDPE, PP and PVC (used in construction). In contrast, PUR, fibres, elastomers, bioplastics, marine coatings and road markings are not recycled, and PS, ABS, ASA, SAN and other polymers can only be recycled to a limited extent (OECD, 2022b). The data on recycled plastics considered as secondary plastics is based on available data on plastics labelled for recycling and also takes into account losses in the process, e.g. when plastics are collected for recycling but cannot be recycled. It is also assumed that biobased plastics or bioplastics are derived from biomass such as corn, sugarcane, wheat or biomass residues and have similar properties to fossil-based plastics (OECD, 2022b). They are considered to be more sustainable due to biomass carbon uptake (Benavides et al., 2020), its inherent energy coming from the sun and its relatively short replacement period.

As recycling rates increase, more plastic waste is collected and processed, leading to a greater supply of recycled plastic material. Consequently, the increased supply can make recycled plastic more available and potentially cheaper, influencing manufacturers to use more recycled plastics in their products. However, the rebound effect, that can be among the outcomes of increased recycling rates, is excluded in this study, as it depends on various factors including economic conditions, regulatory frameworks, consumer behaviour and other (Nordahl and Scown, 2024).

3.3.2. Assumptions and limitations of the method

When projecting plastic quantities to 2050 on the basis of historical data, several limitations of the methods are observed. The projections obtained are based on a single data source (OECD, 2022b). While using a single source ensures consistency across the dataset and facilitates a more streamlined analysis, it may also introduces potential biases and limitations. A single dataset may not capture all of the variability, or nuances present in global trends in plastic volumes, which can lead to potential overlaps. In addition, the scope of our dataset is limited with a relatively small number of data points. This limitation may affect the robustness and accuracy of the projections, as fewer data points may not be sufficient to adequately represent complex and long-term trends. The quality of the data itself may also represent a limitation due to potential data inaccuracy, biases in the datasets and gaps in the completeness of data.

The optimisation models rely on a single variable - time - which can be a straightforward and often useful predictor, but oversimplifies the complexity of factors affecting the plastic quantities. By not accounting for additional variables such as GDP, population growth or demographic factors such as age distribution, the model may miss important correlations and drivers of plastic use. For example, economic growth often correlates with increased plastic consumption, while population growth can influence both demand and waste generation (Lebreton and Andrady, 2019). The ability of the model to capture the multi-layered dynamics of plastic use could result in forecasts that are less accurate or representative of future scenarios.

One notable limitation of the optimisation model is that some coefficients in the model may become zero, thereby eliminating certain terms from the prediction function. This can simplify the model, and it can also lead to a loss of important dynamics that could affect the accuracy of the projections. In addition, the optimisation model used to determine these coefficients can occasionally lead to infeasible solutions or produce unrealistic projections. Such examples are where the model projected a plastic volume of zero in 2050, or projected volume is excessively high in 2050. Based on authors' judgment, these results were excluded and instead opted for projections that better aligned with observed historical patterns. To address these challenges, the projections with different number of coefficients are obtained, initially including as many terms as possible in the optimisation process.

Another limitation is the precision of the coefficients, which were calculated to 15 decimal places. While this precision may improve accuracy, as evidenced by a lower SSR values, it also leads to potential overfitting, where the model may be too sensitive to small variations in the data as discussed in the following. Additionally, there is potential risk of underfitting, which can lead to overly optimistic and unrealistic forecasts, especially when dealing with uniformly increasing variables, as in this study using annual data. To balance the risk of over- and underfitting, prediction functions with different equation terms and effective nonlinear modelling techniques are incorporated into the predictive model. By including a more complex function with more coefficients, the model aims to capture the underlying patterns more accurately and limit noise.

Another limitation of the method could be the structure of the prediction function itself, in particular the equation terms defined. These terms were formulated based on authors' selection regarding the rational expected values of plastic quantities in the future based on historical data patterns. However, this introduces a certain degree of subjectivity, as there may be alternative equations or terms that better capture the underlying trends or dynamics of plastic quantities. In addition, the number of terms included could either be insufficient or too extensive, making the model oversimplified or overcomplicated. This subjectivity in the equation structure leads to uncertainty in the projections, as different formulations may lead to different results. Although the terms selected aim to balance simplicity and accuracy aspects, it is important to recognise that other approaches may offer different insights or levels of precision in predicting future plastic quantities.

A relatively small data set used in this study may contribute to the risk of obtaining inaccurate predictions: 40 data points for plastic use by region, 30 for plastic use by polymer and by application, and 20 for the EoL of waste generated. In addition, the lack of consideration of synergies or antagonisms between years apart from their representation in the historical data is noted as limitation. In the model, the projection for each year is treated independently, without taking into account the potential impact that plastic use in one year could have on the following years. This lack of dynamics between years ignores possible compounding effects or feedback loops that could either strengthen or weaken future trends. However, the annual data points are representative of the underlying socio-economic, technological and policy trends that evolve over time.

Furthermore, the extrapolation extends over a long period up to 2050, i.e. over 30 years, which can lead to considerable uncertainties. Such long-term projections are inherently risky as they attempt to predict outcomes far beyond the range of historical data used for obtaining the correlations. In addition, projection approach in this study is not incremental, but is calculated all at once, which may not capture gradual shifts. This could lead to less accurate projections, especially for the years more far away from the known (historical) data. To address this risk, the optimisation model was tested on several data sets by gradually forecasting the data in steps of ten years. The prediction models first projected the period from 2020 to 2030, where in further steps the predicted values from the first decade were included in the data set to forecast the next decade (2031 to 2040), and similarly for the final period from 2041 and 2050. The results showed that prediction functions using a gradual prediction strategy were consistent with the optimisation approach used in this study, as the deviations were observed only in the third decimal place (up to 0.001 Mt). This proves that optimisation models used in this study provide consistent projections of future plastic volumes.

One of the key assumptions in this study is also that minimising *SSR* as the objective function could ensure effective forecasting by optimising the coefficients of the prediction functions. While minimising the *SSR* helps in fitting the model to the data, it has its limitations, such as the sensitivity to the magnitude of the data as they are measured in the original units and complex and nonlinear nature of the model. To address these limitations, a solver suited for highly nonlinear constraints CONOPT is employed, however it does not guarantee global optimality. Thus, the accuracy and reliability of the model is additionally assessed using several performance metrics such as *MAD*, *RMSE*, and R^2, where R^2 is dimensionless, allowing for comparisons and providing a better indication of the reliability of the model. In addition, a range of possible outcomes and projections is explored and compared to established projections from other sources (see also Table 4 in Results and conclusions). This comparative approach helps to ensure that realistic projections are provided.

3.3.3. Limitations of projecting plastic quantities considering interventions for sustainability

When modelling possible reductions in plastic use and pollution, several limitations should be acknowledged due to various factors such as technological progress, regulatory measures and changes in consumer behaviour. In general, the implementation of different measures to reduce amounts of virgin plastic and plastic waste and to improve waste management includes a wide range of actions and efforts, which can have different outcomes when dealt with in isolation or when different efforts are combined. It should be noted that while this study examines limited number of these efforts or interventions aiming at improved sustainability, each intervention has its own limitations.

In the analysis regarding the different growth rates of plastic consumption, only a decreasing rate until 2050 is assumed and aggregated regional data containing an average consumption rate for all the regions is used, which may not take into account regional differences in plastic consumption patterns. In addition, the assumption of a uniformly declining growth rate may oversimplify the complex interactions between global fluctuations. Potential impacts of unprecedented events such as significant events due to climate change and other effects on the environment, major technological breakthroughs, future pandemics or other influential factors which could dramatically change the trend of plastic consumption in unpredictable ways, is not fully considered.

When forecasting the quantities of plastic packaging based on reduction and recycling targets, one of the key limitations is predicting plastic quantities solely for the packaging sector. While this sector is highly relevant due to its considerable size and its link to single-use plastics, for which there exist numerous targets, policies and efforts, it does not include the full spectrum of plastic use in other sectors. This narrow focus may overlook broader trends in plastic consumption and recycling that could influence overall plastic quantities. In addition, simplified constraint that only up to a certain percentage (80 % was set by authors) of plastic packaging can be recycled reflects practical constraints such as material contamination, inadequate recycling infrastructure and the limitations of current recycling technologies. However, this assumption may not fully account for future advances in recycling technologies or changes in consumer behaviour that could either increase or decrease this percentage.

For the projections of plastic quantities regarding the use of plastics with partial substitution of fossil-based plastics by bio-based alternatives, bioplastics production is limited to the bio-PE, in particular the production route via bioethanol, which is obtained from biomass sources such as sugar cane, corn, switchgrass and forest residues. This focus on a single production pathway ignores other potential methods of bioplastics production that may become viable in the future and limits the adaptability of the model to technological advances. In addition, the calculations of required biomass quantities and corresponding land areas are based on global average data for bioplastics, including land use, yields and conversion rates. This approach may not consider regional differences in agricultural productivity, land availability or different preferences for biomass feedstocks, which could have a significant impact on the feasibility and sustainability of large-scale bioplastics production. Additionally, it is considered that up to 50 % of HDPE and up to 55 % of LDPE could be replaced by bio-based plastics (Brizga et al., 2020). However, technical substitution potential of fossil-based plastics with bioplastics might change due to various factors, such as the share of feedstocks of biological origin which are required for bioplastic production, bioplastic deterioration with time and other.

4. Results and discussion

Historical data on past and recent trends in plastic consumption by region ($q_{\mathrm{m,region}}$), plastic use by polymer type ($q_{\mathrm{m,type}}$), by application ($q_{\mathrm{m,application}}$), and EoL of plastic waste by region ($q_{\mathrm{m,EoL,region}}$) were incorporated into the proposed optimisation approach to identify patterns and predict future plastic use and consumption and EoL management. Besides projecting plastic quantities by using purely historical data, also plastic quantities were projected which additionally consider certain interventions aiming to improved sustainability. The correlations were developed as functions of time that consist of SSR as the optimisation objective and different terms of equations involving different number of coefficients to be optimised.

The resulting NLP models for the obtained correlations of plastic use and waste contain from 96 to 114 single equations and 156 single variables. The number of single equations depends on the number of data points provided (from 20 to 40 data points), while the number of single variables varies depending on the type of equation determined to achieve an optimal fit of the data (all correlations are developed from Eq. (1) that includes 42 coefficients). The optimisation was performed in GAMS using the CONOPT solver, where solutions were obtained on a personal computer with an Intel® Core™ i7-10750H CPU @ 2.60 GHz processor with 16 GB of RAM. All solution times were <5 s, except for the use of HDPE plastics, where the correlation was determined in 13.5 s (see Table C1 in Supplementary material). The model code is available upon request.

The objective values for obtaining the solutions are collected in Table 3, where the consumption of plastic in certain regions is indicated in the subscript, where it replaces the term “region”, plastic use by certain polymer type replaces the term “type”, plastic use by application replaces the term “application”, and EoL of plastic waste by certain end-of-life fates replaces the term “EoL”.

The lowest objective values of *SSR* were obtained for $q_{\mathrm{m,Personal\ care}}$ (0.00001), followed by $q_{\mathrm{m,Littered,global}}$ (0.001), and $q_{\mathrm{m,MarineCoatings}}$ (0.002), while the highest *SSR* obtained was 1434.9 for $q_{\mathrm{m,World}}$. It should be noted that the values of the SSR in the table are given to two decimal places for reasons of clarity and small values (below 0.005) are therefore shown as an approximation to zero (≈0.00).

Apart from the differences between the data, the results reflect the dependence of the *SSR* values on the scale of the data examined, as the calculation of the *SSR* is based on the original units of the data (in Mt/y). As shown also in Figures in Part B of Supplementary material, is this indicated by the high *SSR* values for total use, where a high rate of plastic consumption is reported (see Fig. B1k), and vice versa for example for plastic use of marine coatings (see Fig. B2g), plastic used in personal care (see Fig. B3g), and littered plastic waste (see Fig. B4c).

4.1. Projecting plastic quantities using historical data

The results obtained provide forecasts on plastic consumption by region and on the use of plastics, broken down by polymer types and applications, as well as on the EoL of plastic waste both at global level and for the regions studied. The prediction functions developed consider historical trends in plastic quantities excluding the consideration of additional interventions for sustainability.

4.1.1. Projections of plastic consumption by region and use by polymer type and application

First, plastic consumption for the 10 regions considered (see also Table 2) is projected to 2050 using historical data on the amounts of plastic used annually. The rate of change in plastic consumption by region from 1980 to 2050 is presented in Fig. 1a, indicating that global plastic consumption is expected to almost double within 30 years, reaching up to 832 Mt in 2050. Based on the obtained projections, the amounts of plastic consumed globally (in all the considered regions) are for each year for clarity calculated and presented in Table E1 in Supplementary information (see columns “Plastic consumption by region”). For clarity reasons, the projections obtained for the total plastic consumption are excluded from Fig. 1a as they amount to much higher rates compared to the consumption in each region.

Plastic consumption is estimated to increase in all the regions, especially in the regions China and Other Asia, which are likely to use large amounts of plastic, with the largest volumes being consumed in China (235 Mt/y in 2050). The fastest growth rate of plastic use till 2050 is predicted to be in India, 3.4-times higher compared to 2020. This trend might be attributed to high population growth and GDP, including urbanisation development, economic conditions, degree of industrialisation, and inflation index (Ghayebzadeh et al., 2020), although such factors are not included directly in the correlation but are summarised in the underlying patterns in historical data provided.

The smallest amounts of plastics are predicted to be consumed in Canada, where a slow increase in consumption is observed. This might be attributed to various factors including the implementation of plastic reduction policies and regulations, such as extended producer

Table 3
The optimised values of SSR in Mt for the projection functions obtained[a].

Data set	Sum of squared residuals	Data set	Sum of squared residuals	Data set	Sum of squared residuals	Data set	Sum of squared residuals
Plastic consumption by region		*Plastic use by polymer type*[b]		*Plastic use by application*[c]		*Plastic waste by end-of-life*	
$q_{\mathrm{m,Canada}}$	1.70	$q_{\mathrm{m,ABS,ASA,SAN}}$	0.39	$q_{\mathrm{m,B\&C}}$	29.32	$q_{\mathrm{m,Incinerated,global}}$	2.48
$q_{\mathrm{m,China}}$	130.17	$q_{\mathrm{m,Bioplastics}}$	0.03	$q_{\mathrm{m,C\&IP}}$	9.50	$q_{\mathrm{m,Landfilled,global}}$	11.28
$q_{\mathrm{m,EU}}$	79.60	$q_{\mathrm{m,Elastomers}}$	0.35	$q_{\mathrm{m,E\&E}}$	2.20	$q_{\mathrm{m,Littered,global}}$	≈0.00
$q_{\mathrm{m,India}}$	27.95	$q_{\mathrm{m,Fibres}}$	20.12	$q_{\mathrm{m,I\&M}}$	0.04	$q_{\mathrm{m,Mismanaged,global}}$	5.44
$q_{\mathrm{m,LatinAmerica}}$	5.59	$q_{\mathrm{m,HDPE}}$	10.39	$q_{\mathrm{m,MarineCoatings}}$	≈0.00	$q_{\mathrm{m,Recycled,global}}$	0.22
$q_{\mathrm{m,MEast\&NAfrica}}$	5.77	$q_{\mathrm{m,LDPE,LLDPE}}$	9.72	$q_{\mathrm{m,application,Other}}$	18.95	$q_{\mathrm{m,TotalEoL,global}}$	46.43
$q_{\mathrm{m,OtherAfrica}}$	11.46	$q_{\mathrm{m,MarineCoatings}}$	≈0.00	$q_{\mathrm{m,Packaging}}$	50.42		
$q_{\mathrm{m,OtherAsia}}$	31.50	$q_{\mathrm{m,type,Other}}$	29.80	$q_{\mathrm{m,PersonalCare}}$	≈0.00		
$q_{\mathrm{m,OtherCountries}}$	89.12	$q_{\mathrm{m,PET}}$	1.41	$q_{\mathrm{m,RoadMC}}$	≈0.00		
$q_{\mathrm{m,USA}}$	66.90	$q_{\mathrm{m,PP}}$	21.03	$q_{\mathrm{m,Textile}}$	13.10		
$q_{\mathrm{m,World}}$[d]	1434.91	$q_{\mathrm{m,PS}}$	2.01	$q_{\mathrm{m,Transportation}}$	18.06		
		$q_{\mathrm{m,PUR}}$	1.50	$q_{\mathrm{m,Total}}$[d]	891.58		
		$q_{\mathrm{m,PVC}}$	10.20				
		$q_{\mathrm{m,RoadMC}}$	≈0.00				
		$q_{\mathrm{m,Total}}$[d]	891.58				

Intervention analysis for improved sustainability

Data set	Sum of squared residuals	Data set	Sum of squared residuals	Data set	Sum of squared residuals
Global plastic consumption under varying growth scenarios		*Plastic use in packaging with sustainability targets*		*Plastic use with bioplastic substitution*	
$q_{\mathrm{m},I=1.325}$	217.91	$q_{\mathrm{m,Packaging,reduction,with\ target}}$	113.31	$q_{\mathrm{m,Bio-HDPE}}$	0.75
$q_{\mathrm{m},I=0}$	61.12	$q_{\mathrm{m,Recycled\ packaging,without\ target}}$	≈0.00	$q_{\mathrm{m,Bio-LDPE}}$	0.74
$q_{\mathrm{m},I=-0.5}$	0.06	$q_{\mathrm{m,Recycled\ packaging,with\ target}}$	0.17		
$q_{\mathrm{m},I=-1}$	0.17				

[a] All the data were inserted in the model in the units Mt/y, while the objective values (sum of squared residuals (SSR) values) are provided in the unit Mt.
[b] Abbreviations for polymer types: acrylonitrile butadiene styrene (ABS), acrylonitrile styrene acrylate (ASA), high-density polyethylene (HDPE), low-density polyethylene (LDPE), linear low-density polyethylene (LLDPE), polyethylene terephthalate (PET), polypropylene (PP), polystyrene (PS), polyurethane (PUR), polyvinyl chloride (PVC), styrene acrylonitrile resin (SAN).
[c] Abbreviations for applications: Building and construction (B&C), Consumer and institutional products (C&IP), Electrical/electronics (E&E), Industrial/machinery (I&M), Road marking coatings (RoadMC).
[d] Comparably higher SSR values are obtained for world plastic consumption ($q_{\mathrm{m,World}}$ is 1434.91 Mt) and total plastic use by polymer type and application ($q_{\mathrm{m,Total}}$ is 891.58 Mt), which is due to the high rates of plastic consumption and use, while for plastic consumption by region, an additional reason is the higher number of data points included in the model.

responsibility, as well as increased awareness of the negative impacts of plastic waste on the environment (Diggle and Walker, 2020), which could influence the trends in plastic use. In addition, more emphasis is being placed on sustainable consumption and production practices, resulting in consumers opting for and being willing to reduce single-use plastic packaging personally (Walker et al., 2021). A comparable incremental rate was noticed for the regions Latin America, Middle East and North Africa, and Other Africa, where plastic consumption is predicted to range between 35 and 42 Mt/y. A similar trend is seen in the forecast for the category Other countries, although higher plastic consumption is estimated. For Europe, a slightly negative increment rate was observed from the year 2042 to 2050, suggesting that efforts (which are indirectly integrated into the model as a part of the historical data available) such as the Single-Use Plastics Directive and targets introduced by the EU to reduce plastic use and promote sustainable practices are providing satisfactory results. This aligns with the historical data, which indicates a gradual deceleration in consumption. In addition to the positive projection for the EU, the amount of plastic consumed in the USA will remain nearly constant between 2040 and 2050.

Different plastic types have a variety of functions, and they can be used widely in various sectors. Consequently, the use of all plastic types is predicted to increase in the future, as shown in Fig. 1b. According to the plastic used by polymer types shown in Fig. 1b, PVC and PP are expected to be the most widely used types of plastic in 2050, suggesting that these two types should be the main focus of plastic reduction strategies. PVC is one of the most used types of plastic, as it is convenient in construction, the automobile industry, health care, packaging and other sectors. The use of PVC is predicted to increase 2.5-times in the next 30 years. PVC has experienced significant growth and widespread application in various sectors for several reasons, such as versatility, durability, cost-effectiveness, fire-resistance, and ease of installation. PP has good chemical resistance, is strong, and has a high melting point, making it good for various applications, such as medical and laboratory use, food containers, consumer products, and textiles.

Plastics are used in almost all the sectors, including packaging, building and construction, textiles (e.g. clothing and other products), consumer and institutional products (e.g., goods or services for use by individuals or households, and by organisations, businesses, or institutions), tyres and other transportation applications, electrical and electronics, and industrial and machinery. Due to their versatility, the global use of plastics is increasing in every application (OECD, 2022b), as shown also in Fig. 1c. Especially high increase (2-fold increase between the years 2020 and 2050) is shown for plastic types used in packaging, such as PP, HDPE, and PET, due to the increased demand for packaging materials. A slightly lower increase is observed for the use of other plastic types. Nevertheless, the use of all types is estimated to increase by >1.5-times. A comparable increase is also observed for bioplastic materials. In addition, the projections are obtained for the total use by polymer type/application ($q_{\mathrm{m,Total}}$), however they are not shown in Fig. 1b due to their considerably higher values.

Historical data shows that the strongest increase in total plastic use occurred in the first decade (1990–2000), as quantities almost doubled in 2000 compared to 1990. In the next two decades, i.e. 2000–2010 and 2010–2019, plastic consumption continues to increase, albeit at a slower

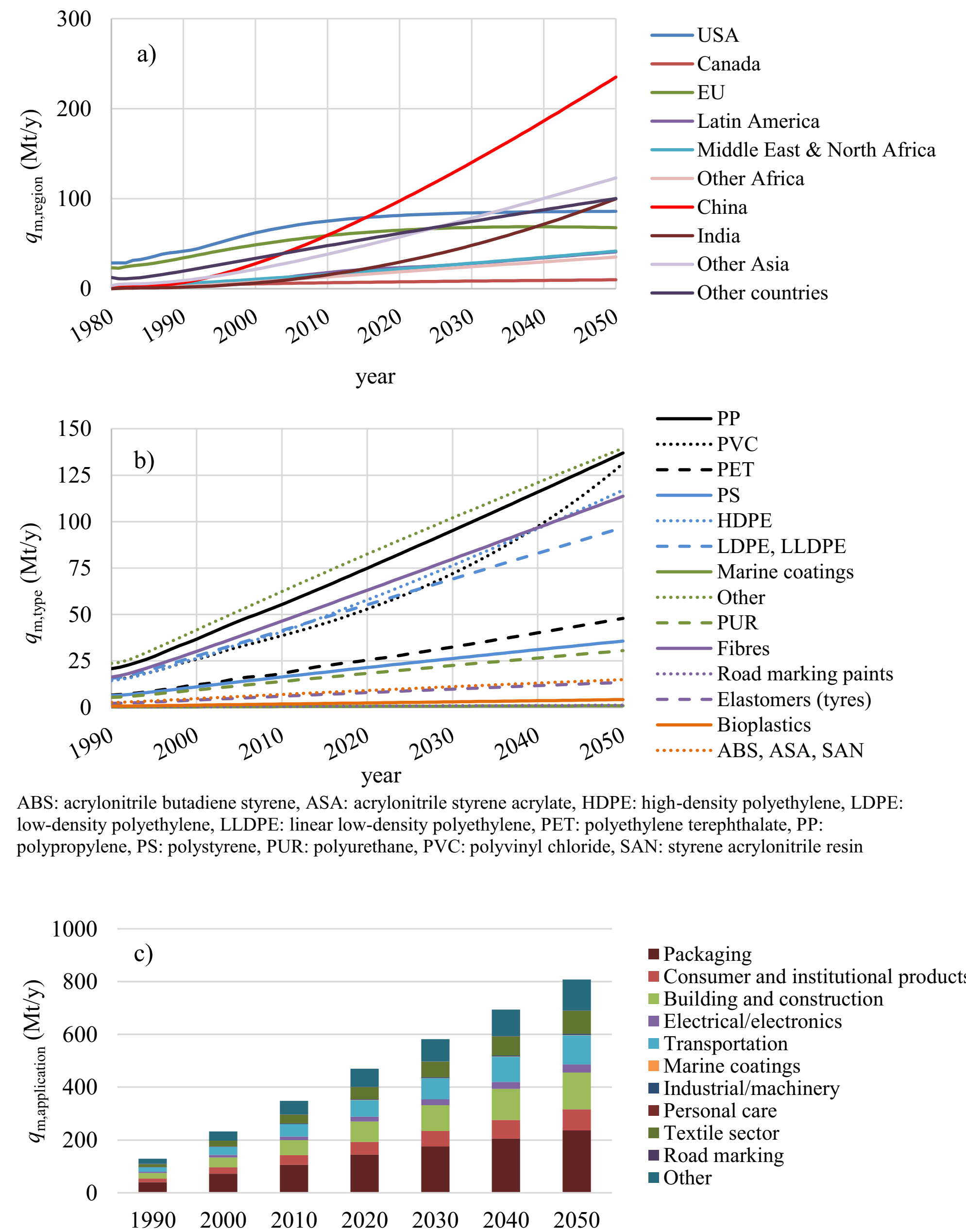

Fig. 1. Projections of a) the plastic consumption by region from 1980 to 2050, b) the plastic use by polymer type from 1990 to 2050, c) the plastic use based on application for decades from 1990 to 2050.

rate, as the rate halved in 2010 as compared to 2000 and more than halved in 2019 as compared to 2010. The observed trend shown in the prediction models (Fig. 1c) indicates a reduced growth rate of plastic consumption in the coming decades.

While the previous surge in plastic consumption up to 2019 is characterized by an average increase of 3.7 times, the forecasts indicate a more modest average increase of 1.7 times for the projected period till 2050. In terms of specific applications, the change in volumes used for consumer and institutional products, buildings and construction, industrial machinery and others is below the average growth rate. The transportation and textile sectors, on the other hand, show an above-average increase. Packaging proves to be the largest consumer of plastics, with volumes rising from around 39.9 Mt in 1990 to 236.3 Mt in 2050, as shown in Fig. 1c. While building and construction, transportation and textile applications also show a significant increase, reaching around 139.1 Mt, 110.6 Mt and 86.3 Mt by 2050 (5.4, 5.5 and 6.9 times the level in 1990), the quantities used for personal care products, marine coatings and road marking are more modest. However, correlations show that despite fluctuations in plastic volumes, the distribution of plastic consumption across the various applications remains relatively stable.

It should be noted that the projection of global plastic consumption ($q_{m,World}$) and the prediction models for each region separately and summed together ($\sum_{region} q_{m,region}$) result in a range of 735.4 Mt to 832.4 Mt in 2050. Similarly, the projection based on the polymer types shows that between 825.8 Mt (obtained by fitting the total use by type, $q_{m,Total}$) and 884.0 Mt (obtained by summing the fitted values of each type, $\sum_{type} q_{m,type}$) of plastic will presumably be used in 2050. As for quantities forecasted for plastic use in application, the projections obtained by the summing of the fitted values of each application ($\sum_{application} q_{m,application}$) suggest that 807.5 Mt will be consumed in 2050, while the projection of the total use by application is the same as the total use by polymer (as they come from the same data sources) resulting in an estimated quantity of 825.8 Mt ($q_{\mathrm{m,Total}}$). As mentioned in Table 2, the data for the total use of plastics by polymer and application presents the identical data set, thus, the projected quantities in 2050 are the same in both categories (825.8 Mt). The overall range of predicted plastic quantities in use is summarised in Table 4 and compared with other studies in the following sections (see section 4.1.3 Comparison of predicted plastic quantities with other studies). In addition, the explanation of the calculation procedure and the data obtained are collected in Part E in Supplementary information.

4.1.2. Projections of plastic waste by EoL in regions

The EoL of plastic waste generated globally for the years 2000, 2020 and 2050 is shown in Fig. 2. It can be seen that the share of mismanaged waste will decrease in the future years. However, because of the increased amounts of plastic waste in general, the volume of mismanaged waste will still be very high. A similar percentage of waste is predicted to be landfilled in 2050 as in 2020 (48.97 % vs 48.72 %), as shown in Fig. 2. This fact still shows some improvements, as developing countries are expected to switch more from dumpsites to sanitary landfills. This conclusion also supports the decreased share of mismanaged plastic waste, which will decrease by 9 % in the next 3 decades. The share of incinerated and recycled waste is estimated to rise by about 4 % in both cases. Overall, this study predicts a plastic waste generation range between 832.8 Mt in 2050, taking into account the historical data of total waste generated ($q_{\mathrm{m,TotalEoL,global}}$), and up to 873.7 Mt in 2050 if the sum of the projections of the individual EoLs in considered ($\sum_{\mathrm{EoL}} q_{\mathrm{m,EoL,global}}$).

Multiple factors contribute to the marginal percentage increase observed in recycling and incineration rates, including the inadequate presence of recycling facilities and incinerators in developing countries, leading to a reliance on landfilling as a primary disposal method. Additional challenges for recycling are difficult sorting conditions and requirements (Chan, 2020), relatively low overall plastic collection rates (Hopewell et al., 2009), and the high cost of current and advancing recycling methods (Smoliński et al., 2022). Another concerning aspect of plastic waste is construction plastic, because those materials are installed for decades before they become waste. After use, some of the construction materials are out-of-date and are harder to incorporate into a circular system (Geyer et al., 2017). This is reflected in on average slightly lower amounts of plastic used globally in 2050 (from 735.4 Mt to 884.0 Mt, see Table 4) compared to quantities of plastic waste generated in the same year (from 832.8 Mt to 873.7 Mt). Although the plastic use is lower in this year, some of the plastics used in the past decades become waste in the years later (in this case in 2050).

In contrast, the cumulative use is slightly higher compared to the amount of cumulative plastic waste produced globally, due to some plastic material remaining in the stock (such as furniture and other consumer goods, electronics, building and construction materials, automotive components and others). Plastic stock is calculated as the sum of the annual differences between plastic consumption or use and plastic waste (see Eq. (E1) in Supplementary information). Its accumulation over the 50-year period is considered (from 2000 until the end of 2050) due to unavailability of the data on plastic waste before the year 2000. Table E2 shows the values for accumulated stock for each year considering different combinations of projections for plastic consumption/use and plastic waste, which are described in more details in Supplementary material. The accumulated plastic quantities until 2050 range between 2062.9 Mt and 4725.0 Mt (around 3500 Mt on average), as presented in Table E3 in Supplementary information.

In addition, the projected values of the plastic waste EoL fate for different regions are presented in Fig. 3. In 2050, it is estimated that more than half of the plastic waste generated will be landfilled in several regions, with the highest proportions in Canada (75 %), the USA (69 %), the Middle East and North Africa (64 %) and Other countries (56 %), of which Middle East and North Africa is the only region where the proportion of landfilled waste is expected to increase. Although landfilling remains the most common disposal method due to land availability and cost-effectiveness (Machado and Hettiarachchi, 2020), the overall trend for landfilling is expected to decrease in several regions. However, for some regions it is expected that landfilling will increase, especially in India, where an increase of 9.9 % (from 36.7 % in 2020 to 46.6 % in 2050) is forecasted. In contrast, the increase is less significant in the rest of Africa (from 30.3 % in 2020 to 32.4 % in 2050) and in China (from 35.9 % in 2020 to 38.3 % in 2050).

In the contrast, the EU is expected to achieve a significant decrease in plastic waste disposed of in landfills, falling to 16 % by 2050. This decline can be attributed to the strict landfill regulations in some EU countries. Accordingly, the share of improperly disposed or mismanaged plastic waste in the EU is expected to decrease rapidly, while the share of incinerated plastic waste is estimated to increase by around 7 % over the next 30 years. Furthermore, the share of recycled plastic waste in the EU is expected to increase by 17 % by 2050. In contrast, recycling rates in developing regions remain low, particularly in the USA (7.2 %), Other Africa (8.6 %) and the Middle East and North Africa (9.2 %). Similar trends can be observed in countries across Asia, Sub-Saharan Africa and South America, where the low recycling rates are primarily due to difficulties in processing plastics (Ferronato et al., 2024).

In terms of incineration, the highest proportions of plastic waste are expected to be incinerated in the EU (51 %) by 2050, followed by China and Other Asia (32 % each). In China, incineration has become more attractive over the last decade due to its effectiveness in reducing waste volumes and recovering energy as a result of the rapid increase in waste generation. However, incineration in China has been criticised for the emissions produced, which may be due to the specific composition of the waste, limited operational experience, insufficient funding for compliance with emission standards and inadequate monitoring measures (Lu et al., 2017). In response, decision makers are now opting for incineration technologies to be based on comprehensive assessments and not just on cost. In addition, improvements in air pollution control systems have been introduced in line with China's new emission standards (Lu et al., 2017).

In 2050, the highest levels of mismanaged waste are projected in Other Africa (58 %), followed by Latin America (33 %) and India (28 %). Many African countries import significant quantities of plastic products and "pseudo-products", which are essentially waste, but there is a lack of up-to-date statistical data and reliable information on plastic production and imports across the continent. This data gap leads to underestimating

Table 4
Summary of past data and projections on global plastic use, waste and stock.

	Past global plastic use, waste and stock (Mt)		
	Use	Waste	Stock[a]
2000	234.0	156.2	77.8
2010	348.9	254.7	948.1
2019	459.7	353.3	1875.3

	Projections of global plastic use, waste and stock for 2050 (Mt)		
	Use	Waste	Stock[a]
This study[b]	735.4–884.0	832.8 − 873.7	2062.9–4725.0
OECD	975.6	798.6	/
Ellen McArthur Foundation	1124	/	/
Geyer et al.	902	1100	/
Stegmann et al.	~1100	~1000	7700

[a] The value stands for the plastic volumes accumulated from 2000.
[b] The one value was obtained by fitting the total plastic volumes for studied set of data (by region, by polymer, by application, by EoL), and the other value by summing the fitted data of each region, polymer type, application or EoL. Consequently, a range of values for a global plastic stock for 2050 is obtained.

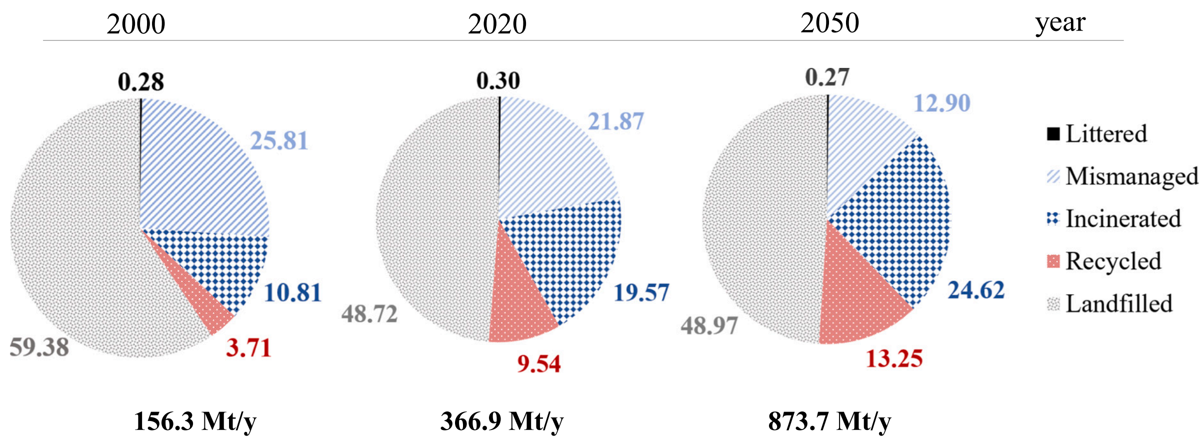

Fig. 2. Projections of the global plastic waste by end-of-life fate in the years 2000, 2020 and 2050.

the problem and often ignored. The growing population and rising living standards in Africa are leading to increased plastic consumption (Moyen Massa and Archodoulaki, 2024).

In contrast, <1 % of the total waste generated in the EU, Canada and the USA is expected to be disposed of improperly. The predicted amount of littered waste remains low in all regions at less than half a percent.

It should be noted that the solutions of the optimisation model for 49 correlations of $q_{\mathrm{m,EoL,region}}$ (combination of 5 EoL and 10 regions, excluding data set for incinerated plastic waste in Other Africa) are based on the Eq. (1) and are consistent with the results and discussions of other projections. Also, the values of the performance metrics show that sufficiently well predictions are obtained for the quantities of plastic waste in the studied regions. The graphical representation of the deviations between the provided data points and the obtained correlation and uncertainty analysis showed that the obtained models are satisfactory.

For clarity and readability, the information on the obtained prediction functions and the optimised coefficient values for 49 correlations for projecting EoL of plastic waste in the different regions studied, are excluded and are available upon request. Also, the results of *SSR*, solution times, R^2, *MAD*, *RMSE* and confidence intervals are excluded for reasons of clarity and readability and are available upon request.

4.1.3. Comparison of predicted plastic quantities with other studies

Due to the increase in plastic use and waste and its environmental implications, this study focuses into past trends in plastic use and waste to forecast future plastic quantities. Overall, the resulting projections indicate that between 735.4 Mt and 884.0 Mt of plastic will presumably be used in 2050. Comparison to other studies, as summarised in Table 4, shows slightly higher values predicted for 2050 in other studies, such as 1124 Mt (Ellen MacArthur Foundation et al., 2017), 975.6 Mt (OECD, 2022b) and 902 Mt (Geyer et al., 2017). Additionally, projections about globally generated plastic waste vary from 798.6 Mt (OECD, 2022b) to 1100 Mt (Geyer, 2020) as shown in Table 4, while this study predicts from 832.8 Mt/y to 873.7 Mt/y of plastic waste generated in 2050. While waste generation is strongly related to plastics for packaging and other short-life products, long-life products, such as building and construction materials, influence plastic stocks. According to Stegmann et al. (2022), up to 7700 Mt of plastic could be accumulated by 2050 and nearly 15,000 Mt by 2100, emphasising the urgency of sustainable management practices. In contrast, estimation in this study indicates that cumulative plastic volumes in stock will increase to up to 4725 Mt from 2000 to 2050. The detailed explanation about obtained ranges of plastic quantities and plastic stock accumulation is contained in Part E in Supplementary information.

Comparing the results of this work with other projections with respect to the global plastic use for 2050, similar trends are observed. Researchers from the OECD predicted a more than five-fold increase in plastics use between the years 2019 and 2060, especially due to strong economic growth in India (OECD, 2022b). A somewhat lower increase was observed in China, where plastic consumption will reach almost a 2.5-fold increase in 30 years. Similarly, an increase in the use of all types of plastic polymers is predicted, among which the most used polymers will be PP, LDPE, and other types that are used mostly in packaging. They estimated that LDPE used in packaging will triple by 2060 as compared to 2019. The use of PP, HDPE, and PET in packaging is also estimated to more than double until 2060. The amount of construction used PVC was predicted to be 2.6-times greater in 2060 compared to 2019 (OECD, 2022b). In another study (Bruggers, 2019), the highest global plastic demand is predicted for materials used in packaging, such as PP and PE, followed by PVC and PET.

Examining other projections on global plastic waste management, researchers from the OECD predicted that global plastic waste will rise from 353 Mt in 2019 to 1014 Mt in 2060, where the share of recycling is projected to rise from 9 % in 2019 to 17 % in 2060 (OECD, 2022b). Geyer et al. (2017) assumed a linear trend in recycling and predicted the global share of recycled plastic waste to take over a great deal at an estimated 44 % in 2050. Furthermore, a similar assumption was made for the global incineration rate, resulting in 50 % of plastic waste being incinerated by 2050.

It should also be noted that for consistency this study considered the data from a single database (OECD, 2022b), while other references contain data sets that show different proportions of plastic use in the sectors, which may be due to different categorisation of the data. For instance, Plastics Europe (2020) reported that 44 % of all plastics are currently used in packaging and 18 % in building and construction, while the OECD (2022b) data for these two categories show 31 % and 17 % of total plastic use.

Numerous other studies have also been conducted on the future of plastic use, and they consistently indicate that without additional policies, there will be a significant increase in plastic use in the coming decades. While the various projections on plastic governance share similarities, uncertainties arise due to the choice of projection methodology and the underlying drivers considered. The estimates for plastic use diverge as we move further in the future. For instance, various research projects suggest that total plastic use will triple by 2050, as shown in Table 4. Studies predict more than a doubling of municipal plastic waste by 2060 (Lebreton and Andrady, 2019), while anticipated doubling of municipal plastic waste from approximately 220 Mt in 2016 to around 420 Mt could even occur earlier, around 2040 (Lau et al., 2020). Further forecasts indicate a more than doubling of mismanaged plastic waste in 15 years, from 31.9 Mt in 2010 to 69 Mt by 2025 (Jambeck et al., 2015), while tripling in 45 years, from 80 Mt in 2015 to 213 Mt by 2060 (Lebreton and Andrady, 2019). Moreover, another

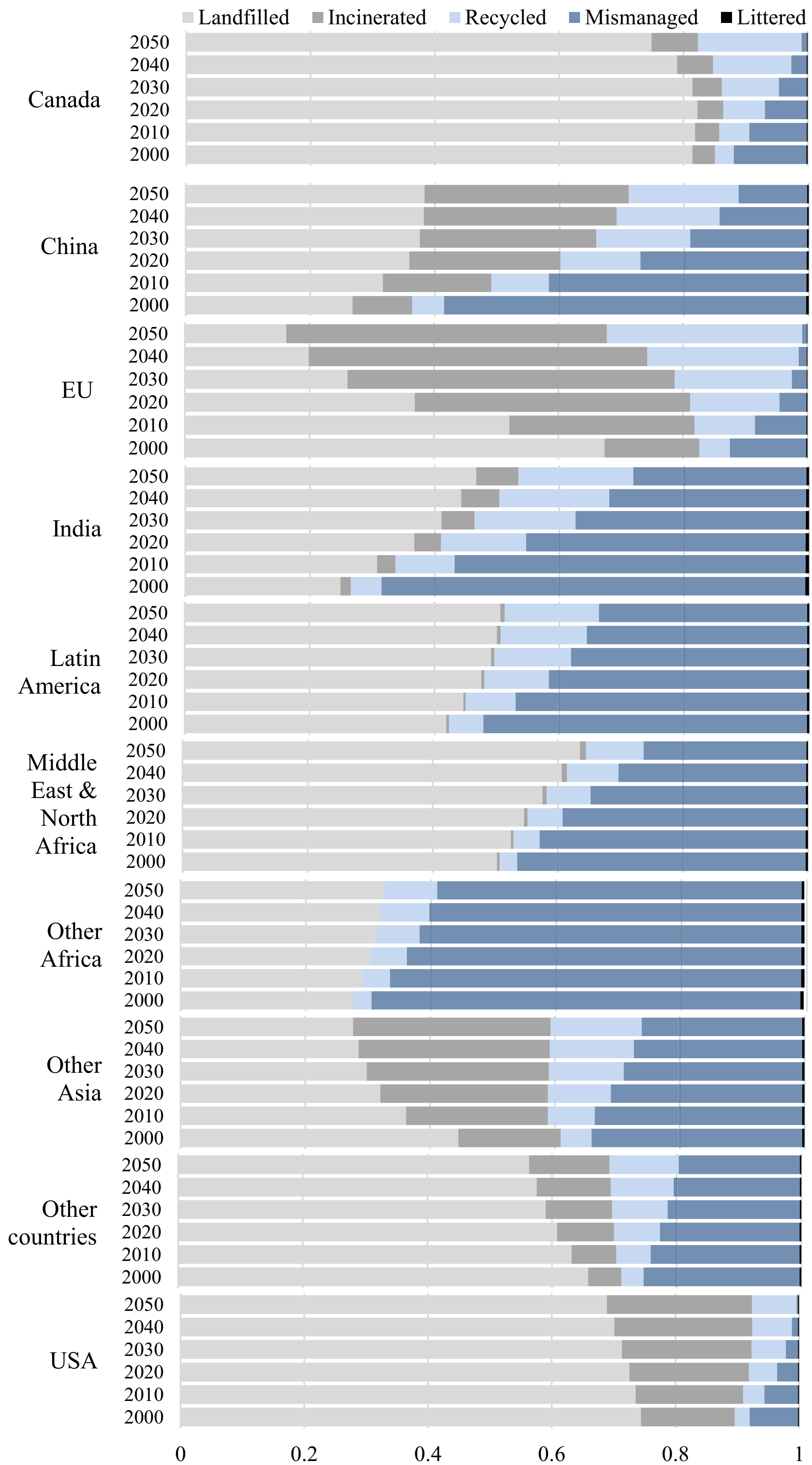

Fig. 3. Contributions on overall results of projections for plastic waste end-of-life for the regions studied.

study (Lau et al., 2020) predicted that mismanaged waste will more than double, from 91 Mt in 2016 to 240 Mt by 2040.

Estimates of plastic use can also be influenced by the approach taken to model plastic waste generation, use, and management. For instance, the OECD study (OECD, 2022b) utilised the ENV-Linkages methodology, which considers all plastic waste, both industrial and municipal solid waste. On the other hand, some other studies solely focused on municipal solid waste (Jambeck et al., 2015). The ENV-Linkages model also refers to a global economic model that captures production, consumption, energy, and trade behaviours across regions, while it links economic activity to environmental pressures, specifically GHG emissions (OECD, 2013). As described by Geyer et al. (2017), the model employs a life-cycle distribution by application to estimate plastic waste.

As it was demonstrated, the projections obtained in this study are in agreement with those reported in other research, indicating consistency with the obtained estimates. Furthermore, the different projections for each category studied such as total amounts and summed projections for regions, polymers, applications and EoL types, resulted in similar ranges of plastic quantities. It should be noted that this convergence of results demonstrates that the methodology is effective in predicting plastic quantities across the categories analysed.

As the results shown in this study as well as most of the existing forecasts, plastic use and the waste generated are expected to increase. Different policies and initiatives have been implemented to curb plastic waste generation, including the plastic (single-use) ban. The significant role of phasing out virgin plastic production has also been mentioned (Simon et al., 2021). However, the exact effectiveness of such an approach needs a better assessment in the future, including the time lag in policy impact and the unintended consequences. Discrepancies between the projected plastic use and actual plastic waste are evident (Table 4). The OECD's projection indicates a waste quantity lower than plastic use, while Geyer et al. (2017) and this study report figures surpassing the use for the same year. It suggests that the prediction approaches have different tendencies and assumptions in capturing the influence on the lifespan of plastic (in use before it ended as waste). Inappropriate or forceful plastic reduction measures may unintentionally lead to behavioural or market changes that counteract the desired or anticipated environmental impact reduction, such as an increase in the waste generation of alternative and less sustainable materials. The durability of materials and suitability for the designated applications should not be compromised. Overall, globally increasing trends in plastic use rates show how critical it is to curb the environmental effects of plastic waste, and to develop longer-term solutions that can lessen the harmful effects of plastics on the ecosystem.

4.2. Projecting plastic quantities considering interventions for improved sustainability

The results of analysing historical trends while considering different interventions towards sustainability show the impacts of the implementation of the new regulations and policies, the potential improvements in the recycling technologies and rates, and the advancements in developing and deploying bioplastics.

4.2.1. Projections of global plastic consumption under varying growth scenarios

The global demand for plastics is increasing, highlighting the role of key countries in the generation of plastic waste. There is an urgent need to reduce demand, improve waste management and introduce plastic substitutes as a global strategy to tackle plastic pollution. Plastic pollution poses a significant health risk worldwide as synthetic plastics degrade slowly and threaten ecosystems. It emphasises the need for all countries to take action to curb the demand for plastics, extend the life of products through repair and reuse, and improve waste management and recyclability (OECD, 2022b).

Here, the potential reduction of plastic use depending on the increment rate of plastic consumption is addressed. The growth rate of plastics for the analysed period is determined based on the projections of plastic consumption by region obtained in the historical trend analysis. According to the historical data, the most recent average increment rate of plastic consumption by region is 2.65 % (around the year 2020), varying from 0.48 % for USA to 5.64 % for India (see Fig. F1 in Supplementary information). Despite increasing plastic quantities, a slow decline in the increment rate can be observed in all the regions over time, reaching the lowest global average value of 1.38 % in 2050, ranging from 0.04 % for USA to 3.04 % for India, which means that the rate of growth of plastic consumption is slowing down.

The potential future trends in plastic consumption are examined through different average global growth rates for the studied time period. In this analysis, the average annual increment rates between 1980 and 2019 are used to develop the correlation to describe the temporal dependence of the plastic growth for the past decades. In the prediction models, the coefficients in correlations are optimised to achieve the best fit between the past increment rates while considering the fixed growth rates integrated for 2050, ranging from −1 % to 2.65 %, exploring the gradual incline/decline in global plastic consumption. The highest value examined (2.65 %) represents the most recent trend in global plastic consumption, as presented above. The second value of the increment rate (1.325 %) investigates scenarios in which the increase in consumption is halved by 2050 as compared to 2020. The assumption of constant plastic consumption in the future is indicated by the third value (increment rate 0 %). In addition, negative increment rates such as −0.5 % or −1 % are considered to examine the decline in plastic consumption in 2050.

Since the first value (2.65 %) follows the most recent trend of plastic consumption with an identical growth rate set in 2050, the development of the correlation assuming a constant growth rate between 2019 and 2050 is not required. For other values of the increment rates, the correlations are obtained in the same way as for the historical trend analysis, meaning they are derived from Eq. (1). For negative increment rates (−0.5 % and −1 %) the prediction model corresponds to Eq. (1), which consists of 42 coefficients, while increment rates (0 % and 1.325 %) are projected with Eq. (1) reduced to 24 coefficients to term involving C_{24}.

As seen in Fig. 4, the investigated increment rates show that if the current growth rate is maintained, plastic consumption will exceed the global consumption in 2050 estimated in the historical trend analysis and will reach 1018 Mt, while the expected quantities of plastic consumed globally are between 735.4 Mt and 884.0 Mt ("baseline scenario"). In contrast, other growth rates evaluated indicate lower consumption. This trend reflects the ongoing gradual decline in the growth rate, which is expected to be between 2.65 % and 1.325 % by 2050. As expected, a gradual transition to a steady, constant level of plastic consumption is observed for a scenario where the growth rate approaches zero. In the cases of negative increment rate in 2050, the sharpest decline in plastic use is seen by 2050.

Given the significant environmental impacts associated with unsustainable plastic production and waste, these scenarios also promote a reduction in GHG emissions. However, such strategies need to be complemented by additional measures, including the decarbonisation of energy systems, the introduction of bio-based plastics and the comprehensive implementation of recycling initiatives (Zheng and Suh, 2019).

4.2.2. Projections of plastic use in packaging with sustainability targets

According to the data on plastic use by application in 2019 (OECD, 2022b), packaging accounted for the largest share (31.0 %), followed by building and construction (16.7 %) and transportation sector (13.5 %). Plastic products used for packaging have generally a short lifespan and are often disposed of in the same year in which they were produced (Geyer et al., 2017). However, only a small fraction of this large potential material stream from packaging materials is recycled, despite significant efforts to promote recycling.

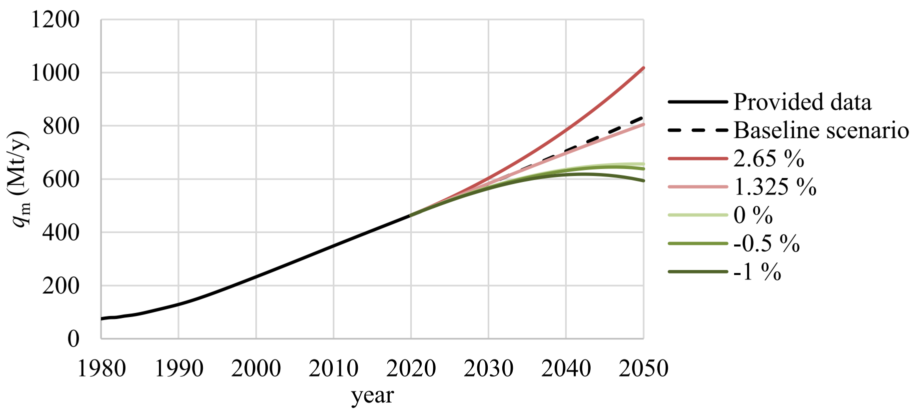

Fig. 4. Global plastic consumption according to the defined increment rate in 2050.

The recycling of plastic waste namely poses a number of challenges due to both the complexity of plastic products and the limitations of current technologies. One of the major problems is the presence of multilayer films and various additives, fillers, pigments and modifiers in commercial plastics. These substances, which are important for the processability and properties of the products, may pose health risks and hinder the recycling process. The variety of substances (>13,000) (European Environment Agency, 2024) and their specific properties can compromise recycling procedures. Thermosets, representing around 12 % of the global plastic production volumes (Morici and Dintcheva, 2022) are at their EoL phase among plastic waste regarded as unrecyclable (Nicholls and Fors, 2024) and usually end up in landfills (Chen et al., 2023).

Another critical challenge is the screening and sorting of plastic waste, a fundamental step in obtaining high-quality recycled materials. The effective separation of plastics from mixed waste streams containing various materials such as different types of plastics, metals, paper and organic residues requires highly sophisticated technologies. Accurate sorting depends on various properties, including magnetic or electrical characteristics, particle size, density, colour and other. The choice of the suitable separation strategy is crucial for the recovery of high-purity polymer fractions (Beghetto et al., 2021). However, it is very challenging to preserve high quality of recycled plastics due to various limitations of both manual and automated sorting and limited accuracy achieved (Pučnik et al., 2024).

In addition, the recycling of hazardous waste (e.g. PVC waste) is complicated by the presence of process additives, such as lead-based stabilizers and certain plasticizers used in materials. Although feedstock recycling is a potential solution for this type of waste, it requires significant investment in industrial infrastructure (Lewandowski and Skórczewska, 2022) and pose significant health risks. For PET waste, chemical recycling processes such as glycolysis and hydrothermal degradations have shown promise. Glycolysis can generate valuable byproducts that serve as raw materials for various applications, including polyurethanes, unsaturated polyesters and plasticizers (Bhanderi et al., 2023). Hydrolytic depolymerisation of PET waste produces terephthalic acid with purity near 100 % (Čolnik et al., 2021). Chemical recycling also offers an opportunity to handle hazardous chemical, contaminated waste, mixed waste, improvements of recycled products quality and other (European Environment Agency, 2024).

However, despite advances in chemical recycling, mechanical recycling remains more established and effective. The lack of adequate infrastructure and technology for chemical recycling and the challenge of replacing current materials with sustainable alternatives continue to limit progress (Beghetto et al., 2021). It should be noted that the displacement of virgin plastics by recycled or secondary ones depends heavily on the technical compatibility of the materials. The compatibility of the secondary plastics cannot be assessed by a single property alone as recycled plastics must meet specific processing characteristics and mechanical properties to be viable in their intended applications (Demets et al., 2021). In addition, the wide variety of plastic types and grades complicates the straightforward identification of technical (un) substitutability, particularly when integrating recycled materials into existing plastic uses (Geyer et al., 2017).

As a large proportion of plastics labelled for recycling were exported to Asia in the past, the import ban imposed by China in 2018 led to a decline in imports of plastic waste to this region (Huang et al., 2020). Following China, other major importers in Southeast Asia have also restricted the import of plastic waste, which has increased pressure on the existing recycling infrastructure and encouraged efforts to shift from downcycling to circular recycling (Vogt et al., 2021).

To limit the use of plastics, 77 countries around the world have introduced regulations that include a complete or partial ban on plastic bags (Statista, 2021). While plastic bag bans are particularly popular in Africa, the majority of the 32 countries that have instead opted to impose a fee (or tax) to restrict the use of plastic bags are in Europe, suggesting that developing countries are more inclined to implement the restrictions as they struggle more with the mismanagement of plastic waste (Statista, 2021). In addition, the European Commission has issued a Directive on single-use plastic, which covers not only plastic bags but also other products such as food and beverage containers, cutlery, plates, straws and stirrers, wet wipes, sanitary products and others, aiming to reduce plastic pollution and marine litter, as single-use plastic items account for 70 % of litter found on European beaches (European Commission, 2019). Furthermore, the draft of the UN international agreement on plastic pollution proposes the implementation of measures to improve the design of plastic products in order to reduce the use of primary plastics, improve the properties of plastics such as safety, durability, reusability, etc. and minimise the release and emission of plastics (including microplastics) (UN, 2024).

Regarding the EoL phase of plastics, the waste management strategies applied differ from country to country. Among the waste management strategies, the application of separate waste collection in Europe resulted in a 13 times higher recycling rate of post-consumer waste compared to a mixed collection system (Plastics Europe, 2022a). Another example is the landfill restrictions introduced in some EU countries (Austria, the Netherlands, Germany, Luxembourg, Sweden, Finland, Belgium, Denmark and Norway) and Switzerland, which resulted in lower amounts of landfilled waste, which amounted to <3 % in 2018 (Plastics Europe, 2020).

However, the implementation of restrictions and bans does not always achieve the desired results. For example, several African countries adopted policies to reduce single use plastic pollution, but enforcement remains difficult and evidence suggests that plastic pollution was not decreased (Adam et al., 2020). Similarly, Sub-Saharan Africa introduced new waste management legislation with the aim to reduce pollution, however after few years, this law has yet to be approved by the relevant authorities (Ferronato et al., 2024). Following China's ban on plastic waste imports, the EU has adopted various strategies and agreements to address the issues of plastic waste, however their effectiveness and impacts are yet to be seen. There is also a significant risk that policies and decisions might have compounding effect, especially for the case of plastic waste, and might lead to increased illegal activities (illegal trade, illegal treatment, landfilling, waste burning), and modifications of shipment routes (Guggisberg, 2024).

As mentioned above, most directives and policies address the reduction of plastic pollution by setting specific targets for packaging. Therefore, the quantities used in packaging in terms of overall waste reduction and increasing the share of recycled plastic are examined and forecasted, taking into account the above-mentioned efforts of the EU and other countries. According to the recycling targets set out in the Packaging and Packaging Waste Directive published by the European Commission, 70 % of all packaging waste should be recycled by 2030, with 55 % of plastic packaging being recycled, counting only material that is converted back into plastics (European Commission, 2018a). Another target, which deals with the overall reduction of plastic packaging, is set for 2040, taking into account a 15 % reduction in plastic packaging waste.

The overall reduction in packaging waste is examined first. As there is no or only incomplete data on the global quantities of packaging waste, the global quantities of plastics used in packaging ($q_{\mathrm{m,Packaging}}$) are used. The values of the correlation coefficients are determined taking into account a 15 % decrease in plastic packaging volumes in 2040 compared to the volumes generated in 2018. The prediction model is obtained using Eq. (1), where 42 coefficients are considered (see section 3 Methods).

The comparison between the forecasted plastic quantities used in packaging with and without the reduction target is illustrated in Fig. 5. Applying the reduction target, the amount of plastics used in this sector decreases from 138 Mt in 2020 to 100 Mt in 2050, which means a 27.3 % decrease in plastic packaging for the projected period. If, on the other hand, the use of packaging is assumed to be predicted on the basis of historical data without additional measures, packaging volumes would increase to 236.3 Mt in 2050.

In the second part, the analysis focuses on the proportion of packaging waste that is recycled. Data on the global use of plastic packaging ($q_{\mathrm{m,Packaging}}$) is considered and projections of the historical trend analysis are used to describe the quantities used for this application. Based on the quantities of global plastic waste management, the proportion of recycled plastics compared to total waste is determined. The share of recycled plastics was increasing from 3.7 % in 2000 to 9.3 % in 2019 and is expected to reach 55 % in 2030 in accordance with the target proposed in Directive (European Commission, 2018a), denoting that more than half of all plastic packaging should be recycled by 2030. As the data on the use of plastics by application is available for the period 1990 to 2019 and the data on EoL is available for the period 2000 to 2019, the data for the period 2000–2019 is considered and the potential impact of the target by 2050 is examined.

It should be noted that recycled plastic waste also includes materials used not only for packaging but also for other applications, such as the recycling of plastics in construction (e.g. window profiles, pipes) (Plastics Europe, 2018). However, the main source of recycled plastics is currently packaging. In Europe, for example, packaging accounts for up to 80 % of all recycled plastic waste (Plastics Europe, 2019). Although comprehensive data on recycled material from other applications in other countries is lacking, it is reasonable to extrapolate this trend globally and assume that around 80 % of all recycled material comes from plastic products used in packaging. In addition, as some plastic packaging is not recyclable for various reasons, an assumption that 80 % of all packaging being recyclable is considered.

The historical and forecasted data on plastic packaging is summarised in Fig. 6, which compares the projections of plastic use in packaging (red curve), the share of packaging recycled following the historical trend analysis (blue curve) and the target of 55 % of all packaging material being recycled by 2030 (black curve). In order to achieve the best fit between the data provided and simultaneously satisfy the recycling target, a correlation is developed using Eq. (1) (see section 3 Methods). The prediction model estimates that by following the recycling target up to 189 Mt of plastic packaging will be recycled in 2050, after the proportion of recycled packaging has strongly increased from 2020.

However, as seen from Fig. 6, by following the current trend, only 34 Mt of plastic packaging will be recycled in 2050 (blue curve), compared to 236 Mt that is forecasted taking into account the recycling target (black curve). It should be also noted that in reality the recycling of all plastic packaging in 2050 is unlikely to be either practical or economically feasible. However, it is expected that the realistic packaging amounts will be between the projected amounts obtained, taking into account improved waste management methods and recycling technologies.

The production of secondary plastics is influenced by the interplay of three key factors, taxation, production and demand. The taxation of plastics, which is intended to curb demand, can reduce the use of secondary plastics. However, the production of secondary plastics depends on scrap from recycled plastic waste, which is increased by measures to promote recycling. The less plastic waste is generated, the less scrap is available for secondary production. These factors work together like a 'push-pull' mechanism: targets for the proportion of recycled content increase the demand for scrap (pull), while recycling measures increase

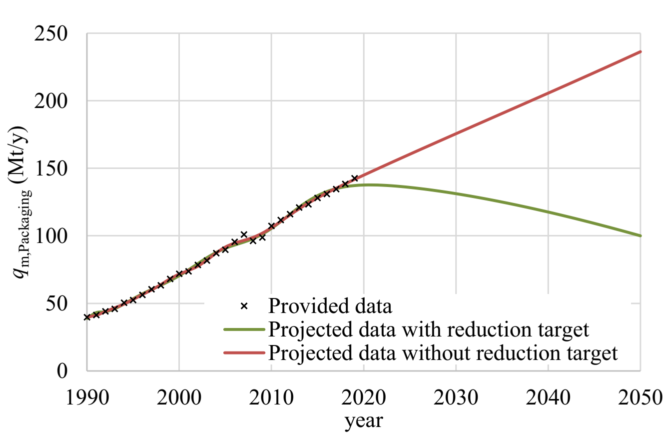

Fig. 5. Projection of plastic use in packaging with and without consideration of the reduction target.

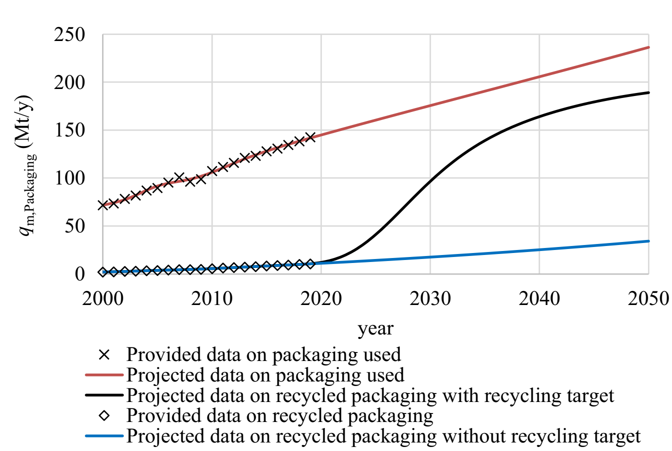

Fig. 6. Projections of plastic use in packaging with and without consideration of the recycling target.

the supply (push) (OECD, 2022b).

4.2.3. Projections of plastic use with bioplastic substitution

The gradual substitution of fossil-based plastics with bioplastics represents a significant shift towards more sustainable materials. Bioplastics derived from renewable biological resources promise to reduce dependence on fossil fuels and reduce associated GHG emissions. The majority of bio-based plastics are produced from sugarcane and corn, while second-generation bio-based feedstocks such as switchgrass and forest residues offer advantages in terms of cultivation methods and/or limited competition with the food processing industry (Mendieta et al., 2020).

Incentives for the adoption of bioplastics come in various forms, including policy measures such as subsidies for bioplastics and taxes on fossil alternatives. The study presented how taxing fossil plastics while subsidising bioplastics can make the production of bioplastic more economically viable. In addition, investments in technologies play a crucial role in increasing the productivity of agricultural feedstocks and reducing land requirements. Technological advances in biorefineries, especially in the use of non-food feedstocks such as algae or agricultural residues, are crucial to minimise competition with food production and improve overall sustainability (OECD, 2022b).

Bioplastics have notable benefits, but also challenges compared to traditional fossil-based plastics. For example, the production of bio-based HDPE has a 15 % lower impact on climate change compared to fossil-based HDPE, mainly due to the lower GHG intensity of bioethylene production. In addition, bioplastics support a circular economy through improved recycling potential. Chemical recycling of bio-based HDPE has been shown to be more effective in reducing climate change impacts than mechanical recycling (Helmes et al., 2022). Despite these benefits, the expansion of bioplastics production comes with significant environmental challenges. The increased demand for land to grow feedstocks for bioplastics can lead to indirect land use effects, such as deforestation, which can result in significant GHG emissions, biodiversity loss and other environmental problems such as eutrophication and acidification. These indirect effects can potentially offset the climate benefits of bioplastics. Furthermore, the production of bio-based plastics often involves the use of additives which contribute to additional GHG emissions and environmental impacts, which complicates the overall sustainability profile of these materials (OECD, 2022b). Competition for land, water and other resources is another major problem. If the production of bioplastics expands, it may compete with food production for agricultural land, which could drive up food prices and affect food security. The cultivation of raw materials also requires significant water and land resources, which could exacerbate environmental pressures, especially in regions already facing scarcity (Brizga et al., 2020).

As the production of bio-PE is one of the most mature and developed technologies, and, as of now, the followed approach is synthesis of the ethylene monomer by dehydrating the bio-ethanol (Siracusa and Blanco, 2020) obtained from bio-based feedstocks. Here, the production of bioethanol is investigated from each bio-based feedstock separately, i.e. sugarcane, corn grain, switchgrass and forest residues. The conversion of biomass into bioethanol through five processes, i.e. biochemical conversion, dry-grind process, gasification and catalytic synthesis, gasification and syngas fermentation, and anaerobic fermentation, is assumed (Čuček et al., 2014). Bioethanol is then used to produce bio-HDPE or bio-LDPE assuming a conversion factor (Benavides et al., 2020).

The bio-HDPE and bio-LDPE plastics produced from bio-based feedstock are characterized as bio-based, non-biodegradable plastics and have the potential to substitute fossil plastics by 50 % in the case of HDPE and 55 % in the case of LDPE (Brizga et al., 2020), and even up to 70 % in the case of PE in general (Zheng and Suh, 2019). The gradual transition from fossil to bio-based PE is projected by developing the correlation of bio-HDPE and bio-LDPE production with the set share of bio-based plastic up to 50 % for bio-HDPE and 55 % for bio-LDPE in 2050. Due to the incomplete data on bioplastics in terms of the exact proportion of specific biobased polymers, the data on bioplastics is used as representative of bio-PE quantities as one of the most popular bioplastics, assuming that the majority of bioplastics are of this type and neglecting other types of bio-based plastics. Based on the similar current use of polymers HDPE and LDPE (56 Mt and 54 Mt in 2019), it is assumed that half of the bio-PE is used for bio-HDPE and half for bio-LDPE production, meaning that data for bioplastics are equally divided for the production of these two polymers. The correlation of bio-HDPE and bio-LDPE is developed according to Eq. (1) (see section 3 Methods). The quantities of fossil-based HDPE and LDPE are then determined by subtracting the bio-based volumes from the global plastic use of these polymers estimated in the historical trend analysis.

The projections of HDPE and LDPE production are shown together with the estimated plastic use in the historical trend analysis in Fig. F2 in Supplementary information. Fossil-based HDPE and LDPE use is projected to initially increase to 76 Mt and 66 Mt between 2035 and 2040 and then decline to 58 Mt and 43 Mt in 2050. The trend for the use of both polymers is similar, with the difference that the amount of bio-based LDPE exceeds the amount of fossil LDPE in the last two years of the forecasted horizon, as defined by the restriction in 2050.

Based on the estimated plastic volume of these polymers by 2050, the quantities of biomass needed to sustain production are calculated and consequently the land area required for the cultivation or harvesting of bio-based feedstocks is determined (see Fig. 7). To produce the same amount of plastics, the amount of biomass required varies from >1477 Mt of sugarcane to 316 Mt of corn grain. In contrast, the smallest area is required for the production of sugar cane, while the largest area is considered for the obtaining forest residue, i.e. 551,429 km^2 and 505,714 km^2 to cover the demand for bio-HDPE and bio-LDPE. The explanation lies in the yield per hectare of the feedstocks, i.e. feedstocks with a high yield per hectare require less land for cultivation or harvesting.

It is important to note that the yields of biomass vary significantly, both spatially and temporally. For example, the available quantity of forest residual biomass that can be collected and used depends on various factors, such as terrain characteristics, the road network for the collection and transportation of woody materials (Qian et al., 2024), and the wood and harvesting system (Hall and Evanson, 2007). Forest residues are generally logging residues for commercial purposes and thinning residues for forest management (Qian et al., 2024), with a wide range of yields reported (Hall and Evanson, 2007). Previous studies estimated different ranges of the amounts of forest residues, some examples are: 6.6×10^{-3}-1.8 t/ha (Fu et al., 2020), 0–3.98 t/ha (Cozzi et al., 2013), 3.8–25.3 t/ha (Carrasco-Diaz et al., 2019), 15–30 t/ha (Eker et al., 2017), and 50–100 oven dry t/ha (Hoyne and Thomas, 2001).

In addition, agricultural yields of other crops are influenced by technological factors such as agricultural practices and management

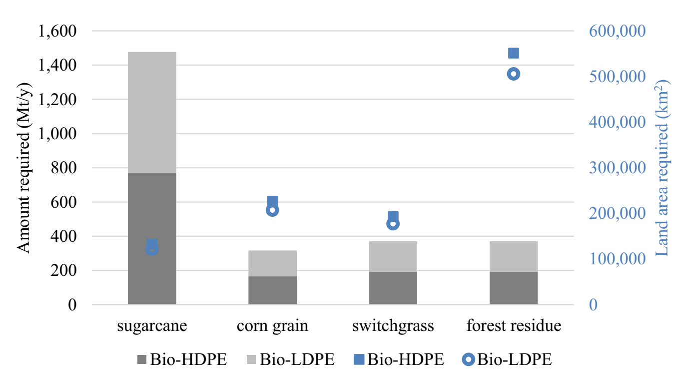

Fig. 7. Estimated amounts of biomass and land area required for bio-high-density polyethylene (bio-HDPE) and for bio-low-density polyethylene (bio-LDPE) production in 2050.

decisions, biological characteristics such as diseases, insects, pests and weeds, and environmental conditions such as climate, soil fertility, topography and water quality (Tandzi and Shelton, 2020), which can lead to significant differences in yield value (Winchester and Reilly, 2015). Significant differences in yields were reported also for sugarcane due to its comparably high yields, while less significant differences were obtained for corn grain and switchgrass. Some examples of sugarcane yields are: 70.8 t/ha (Desalegn et al., 2023), between 52.5 and 121.8 t/ha (Ram and Ramaiyan, 2021), average worldwide yield between 50.3 t/ha in 1961 and 73.7 t/ha in 2022 (Our World in Data, 2024) and between 236 and 280 t/ha (experimental plots in Brazil, (Duke, 1983)). In this study, the following annual yields are considered: 58.15 t/ha for sugarcane, 7.3 t/ha for corn grain, 10 t/ha for switchgrass, and 3.5 t/ha for forest residue (Čuček et al., 2014).

Transitioning from fossil to bio-based plastics reduces the consumption of fossil feedstock such as crude oil and natural gas, which helps to reduce GHG emissions. However, other environmental impact indicators are strongly influenced by the harvesting and cultivation of biomass. Furthermore, it is assumed that decarbonising the energy system is an economically more favourable option for reducing GHG emissions than the use of bio-based plastics (Zheng and Suh, 2019).

5. Conclusions

In this work, a model was developed for predicting the amounts of plastics likely to be used and the amount of plastic waste generated considering purely historical trend analysis and additionally certain interventions aiming at reduced environmental burdens (sustainability constraints). The forecasts from the model can support decision-making to develop mitigation strategies and regulations, such as reducing single-use plastics or improving the recycling infrastructure. The approach developed and used in this study enables making projections based on the inherent dynamics and fluctuations observed in the historical data while indirectly accounting for the multitude of factors that have influenced plastic quantities in the past.

All the projections in this study assume that the global plastic quantities will gradually increase to 2050. They will increase from 464 Mt in 2020 to the range of quantities in 2050 to be between 735.4 Mt/y when considering the world's plastic consumption, and 884.0 Mt/y in 2050 when summing the projections for each polymer type. In overall, the projections show that the amounts of plastic waste will double by 2050, despite expected improvements in recycling rates. To account for the proactive policy measures to tackle the challenges of plastics production and waste management, the intervention analysis for improved sustainability highlights the potential positive impact of such interventions, while complementary measures such as the decarbonisation of energy systems, the production of bio-based plastics and plastics from power-to-X and comprehensive implementation of recycling initiatives are essential to be in line with broader sustainability goals. In addition, the analysis of the projected plastic waste in different regions considering various EoLs and the intervention analysis showed that strict environmental policy measures such as the ban on single-use plastics, plastic taxes and extended producer responsibility may have a significant impact on reducing plastic waste. More strict policies, aiming at reducing environmental burdens together with industrial innovations, show that such countries and regions are more successful in reducing plastic waste, providing a model for other regions. The data shows the importance of continued policy development, industry collaboration and consumer education to further improve sustainability efforts.

The study emphasises the need for interpretable and valid projections to inform environmental policies and to identify future trends in energy and resource use, environmental impacts, and plastic waste scenarios. In order to transit towards more sustainable systems, they must also be adaptable to changing circumstances. The proposed forecasting approach based on historical data offers high interpretability of temporal patterns, seasonal patterns, and event identification and shows autocorrelation. However, the approach is subjected to limitations of extrapolation and potential external influences, especially when forecasting far into the future. In addition, it is important to recognise that while the method used provides valuable insights and predictions, it is not designed to capture the precise influence of specific external factors in isolation. Instead, it uses the cumulative effect of these factors on plastic quantities as reflected in the available data, providing a useful tool for forecasting trends in plastic quantities over time. The proposed prediction model could be improved by integrating alternative indicators and additional variables that could provide stronger correlations with outstanding events influencing plastic flows in the past. Nonetheless, it is crucial to exercise caution in order to prevent potential under- or overfitting when introducing these refinements. By applying mathematical transformations, capturing interactions, or incorporating lagged variables, the predictor functions can be improved to capture time dependencies more accurately.

In the future studies, several perspectives may be considered, including more drastic changes that can result from stronger policy measures or technological developments. Specific points of interest include i) Comparison of different predictive modelling techniques, including machine learning, to identify biases or assumptions in different methodologies, ii) Analysis of plastics trade on a broader scale encompassing a region-specific investigations, iii) The development of bioplastics, especially biodegradable plastics, iv) The development of materials that are easier to recycle through design-for-reuse and design-for-recycling, v) The development of plastic production power-to-X technologies using surplus electricity from variable production of alternative electricity, vi) Better sorting, separation and segregation through a smart waste management system and vii) The development and application of innovative recycling concepts. This is expected to contribute to the transition to more sustainable plastic value chains and reduce environmental footprints.

CRediT authorship contribution statement

Monika Dokl: Writing – original draft, Visualization, Software, Methodology, Investigation, Formal analysis, Data curation, Conceptualization. **Anja Copot:** Writing – original draft, Methodology, Investigation, Data curation, Conceptualization. **Damjan Krajnc:** Writing – review & editing, Visualization. **Yee Van Fan:** Writing – review & editing, Visualization. **Annamaria Vujanović:** Writing – review & editing. **Kathleen B. Aviso:** Writing – review & editing. **Raymond R. Tan:** Writing – review & editing. **Zdravko Kravanja:** Writing – review & editing. **Lidija Čuček:** Writing – review & editing, Supervision, Software, Funding acquisition, Conceptualization.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Appendix A. Supplementary data

Supplementary data to this article can be found online at https://doi.org/10.1016/j.spc.2024.09.025.

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